

Firm Heterogeneity and Racial Labor Market Disparities^{*}

Caitlin Hegarty[†]

Williams College

June 2, 2026

Abstract

When unemployment rises, the change in employment is more severe for Black workers than white workers, even adjusting for differences in industry and other factors. I show that Black employment adjusts more at large firms in particular. Motivated by this evidence, I develop and calibrate a search model with flexibly-specified racial disparities in hiring that vary with firm size. Large firms employ more Black workers on average and reduce Black employment more in a slack labor market, consistent with the data. The model shows that distortions in matching amplify the effects of discrimination, which worsen as firms become more selective.

JEL: E24, J63, J71, M51

Keywords: Racial employment gap, firm heterogeneity, search and matching

^{*}I am grateful to Gadi Barlevy, Charlie Brown, Marcus Casey, Jason Faberman, Bart Hobijn, Luojia Hu, Leticia Juarez, John Leahy, Claudia Macaluso, Erin Markiewicz, Leonardo Melosi, Pablo Ottonello, Tyler Radler, Matthew Shapiro, Mel Stephens, Burak Uras, Toni Whited, and Guanyi Yang for helpful conversations and comments. I also thank various seminar participants and conference participants at UM-MSU-UWO Labo(u)r Day, BSE Summer Forum Macroeconomics of Labor Markets, NBER Macro Perspectives, Liberal Arts Macro, ASSA San Antonio, and Society of Labor Economists for helpful comments. This material is based upon work supported by the National Science Foundation Graduate Research Fellowship Program under Grant No. DGE 1256260. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author and do not necessarily reflect the views of the National Science Foundation.

[†]Caitlin Hegarty (ceh10@williams.edu): Williams College, Department of Economics.

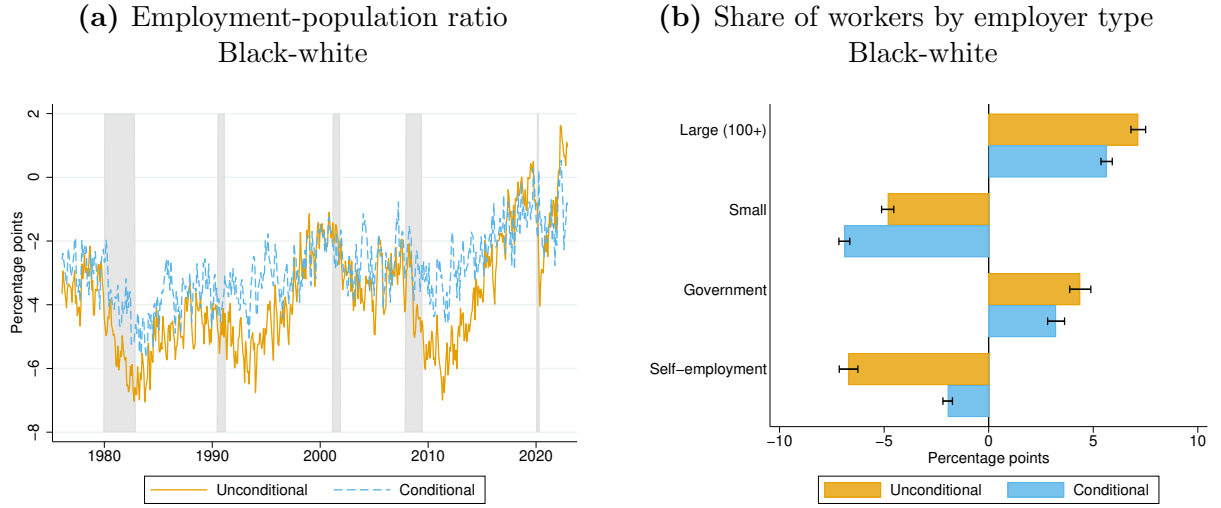
1. Introduction

The Black population in the U.S. faces persistently lower rates of employment than the white population. Additionally, Black employment responds more to macroeconomic conditions, rising more during expansions but also falling more during contractions. For example, over the peak to trough of the Great Recession, the Black employment rate fell by 4.5 percentage points whereas the white employment rate fell by 3.2 percentage points. Observable worker characteristics are unable to explain either the level of the employment gap or its comovement with aggregate conditions, as shown in Figure 1a. Understanding differences in exposure to aggregate labor market risk is important both for addressing persistent racial economic disparities and also for designing equitable stabilization policies in response to downturns.

This paper explores the role of firm heterogeneity and disparities in hiring for explaining the greater sensitivity of Black employment to macroeconomic conditions. Black workers are about 5.6 percentage points more likely to be employed by large firms than white workers, conditional on observable characteristics such as industry and location, as illustrated in Figure 1b.¹ Motivated by these differences in composition, I document new facts about the correlation between aggregate unemployment statistics and individual-level employment transitions by race and firm size. When local unemployment rates rise, Black workers are less likely to be employed, with most of the adjustment coming from large firms. I show that these patterns are consistent with a model in which Black and white workers are ex-ante identical, but Black workers face additional barriers in hiring. Intuitively, these barriers lead firms to favor white workers when aggregate productivity falls. A more nuanced result is that this additional adjustment is not proportional to firm size. Although racial hiring barriers are more severe at small firms, Black employment at large firms adjusts more in the calibrated

¹Mean employment at firms with 100 or more employees for Black workers is 51% in the March supplement to the CPS and 52% in the SIPP.

Figure 1: Racial gaps in employment and employer composition



The solid line in Panel (a) reports the difference in employment-population ratios between Black and white populations over the population 20 and older in the CPS. The dashed line reports the unexplained gap from a Oaxaca-Blinder decomposition estimated separately in each period, with controls for age; age-squared; gender interacted with marital status; education; state; metro area size; modal industry and occupation. Panel (b) reports the difference between the composition of Black and white employed workers, using the March supplement to the CPS. The blue bars report the residual gap after conditioning on the same set of controls as Panel (a) with fixed effects for year. Large firms have 100 or more employees.

model to a deterioration of aggregate productivity. The adjustment of Black employment at large firms is amplified by distortions in matching rates and spillovers between large and small firms. This has two relevant takeaways for policy. First, policies that aim to reduce discrimination by regulating the largest firms may be less effective if the spillover effects from small firms are sufficiently large.² Second, policies that target small firms may be less effective at stabilizing employment for the Black population due to both composition effects and the importance of fluctuations at large firms.³

I use micro-data from the Survey of Income and Program Participation to show that

²For example, firms with 100 or more employees (or government contractors with 50 or more employees) must report the composition of their workforce by occupation to the Equal Employment Opportunity Commission (EEOC) each year.

³For example, the Paycheck Protection Program incentivized employee retention at small firms during the Covid-19 recession.

Black employment and hiring are more sensitive to labor market conditions, even within individuals. When the state-level unemployment rate is 1-standard-deviation higher, Black workers are an additional 0.6 percentage point less likely to be employed than white workers, controlling for worker fixed effects. This change represents about 8% of the unconditional mean employment gap and is driven by the change in employment at large firms. I observe similar patterns in total hiring and job-finding from nonemployment as well, though not in separations.

The empirical patterns show that Black workers are more likely to work for large firms and that large firms contribute more to the worsening of the employment gap when the labor market is weak. I next develop a model to study how racial disadvantages in the hiring process across firms contribute to these patterns. I start with a canonical random search model and introduce three main ingredients: endogenous firm size, uncertain worker productivity, and racial disadvantages in hiring. The first two ingredients create a trade-off for firms between recruiting intensity and selectivity. If firms choose a high selectivity strategy, they pay high search costs but the workers they recruit are very likely to be productive so the cost of turnover is lower. I assume that small firms face higher search costs than large firms, which produces higher turnover rates at small firms.

The third ingredient is the assumed disadvantage for Black workers in hiring. Any assumption will undoubtedly fail to capture the full set of factors that differentially affect labor market outcomes for Black relative to white workers, such as such as employer prejudice ([Charles & Guryan, 2008](#)), wealth inequality ([Derenoncourt *et al.*, 2024](#)), and many more. In the model, I allow for rich interactions among three sources of racial distortions and I use the calibration to discipline the relative magnitudes of each channel. (1) I allow firms to receive noisier signals about Black applicants, particularly at small firms, meant to capture statistical discrimination. This channel could be microfounded with differences in professional networks or differences in monitoring practices which can lead to statistical dis-

crimination as in [Cavounidis *et al.* \(2024\)](#). (2) I allow firms to have utility costs associated with employing workers by race, meant to capture taste-based discrimination. This channel would be consistent with small firms being more influenced by taste preferences of management and facing less legal scrutiny from the EEOC, enabling them to discriminate more. (3) I allow the composition of matches to be distorted by the composition of incumbent employment. This acts as a co-racial hiring channel ([Miller & Schmutte, 2023](#), [Sorkin, 2025](#)). Each of these channels leads to lower employment for Black workers and sorting towards large firms. With statistical discrimination, firms find fewer Black workers who meet their productivity criteria, whereas with taste-based discrimination they find fewer Black workers who meet their inflated standards for hiring. Incumbency bias in matching rates amplifies either form of discrimination.

Finally, I calibrate the model to match moments from the micro-data. I take moments on employment gaps, job-finding rates, and separation rates by firm size directly from the regressions in Section 2. Using the calibrated model, I shut down each hiring friction one at a time to evaluate the importance for the racial composition of employment. I find that statistical discrimination is most important for reproducing the employment gap, while incumbency bias in matching rates is most important for sorting patterns. I use the model to simulate a counterfactual, low aggregate productivity economy, meant to reflect a 1-standard-deviation increase in state-level unemployment from the data. I calibrate the size of the decrease in aggregate productivity such that the change in white employment matches the regression coefficient from the data.

The model predicts that the racial employment gap worsens by 1.21 percentage point in the lower productivity steady state, roughly double the decrease from the empirical section of 0.61 percentage point. Qualitatively, the model captures that the worse employment gap is driven by large firms. Decreased demand for labor leads to a decrease in market tightness, allowing firms to attract more applicants per unit of search intensity. Thus, firms substitute

more towards a high-selectivity strategy, which reduces the surplus they share with workers in wages. This has an outsized effect on employment for Black workers because the increase in selectivity is compounded by discrimination and the feedback effect of reduced wage pressures leads to less demand for Black workers at large firms in particular. The source of racial disparities matters for both the size of the change in the employment gap and the degree to which it is driven by large firms. The employment gap worsens most in a scenario with strong taste-based discrimination, while large firms are most influential when Black workers are more exposed to them in the matching process.

Related literature This paper contributes to several major strands of the literature. First, there is an extensive literature studying the excess sensitivity of Black employment to macroeconomic conditions ([Couch & Fairlie, 2010](#); [Hoynes *et al.*, 2012](#); [Cajner *et al.*, 2017](#); [Aaronson *et al.*, 2019](#); [De *et al.*, 2021](#); to name a few). My finding that the job-finding margin is the most responsive for Black workers during downturns is consistent with recent evidence by [Forsythe & Wu \(2021\)](#) and [Kuhn & Chanci \(2021\)](#). Another related area of research considers the effects of monetary policy on racial inequality (see [Bartscher *et al.*, 2021](#); [Lee *et al.*, 2025](#); [Bergman *et al.*, 2020](#); [Thorbecke, 2001](#); [Carpenter & Rodgers III, 2004](#); [Zavodny & Zha, 2000](#)). This literature demonstrates the policy interest of understanding how racial differences evolve over the business cycle. My paper contributes to this literature by introducing the role of employer heterogeneity, which is interesting on its own for understanding how shocks permeate through the economy, but also could have important implications for economic policies that interact with the firm size distribution.

Second, there is a large literature documenting racial disparities and discrimination in the labor market (see [Lang & Lehmann \(2012\)](#) for an overview). The fact that Black workers are more likely to be employed by large firms was documented by [Holzer \(1998\)](#) and has more recently been emphasized by [Miller \(2017\)](#), [Miller & Schmutte \(2023\)](#), and

Sorkin (2025). The correspondence study literature has shown empirical racial disparities by firm size. Kline *et al.* (2022) finds that callback gaps on the basis of perceived race are prevalent even among the largest employers, though the gaps are somewhat smaller than those found in other studies, such as Bertrand & Mullainathan (2004). Banerjee *et al.* (2018) also finds less discrimination among larger employers in Canada. Empirically, Boulware & Kuttner (2019) finds that discrimination charges are countercyclical. My paper builds on these studies by incorporating heterogeneous discrimination by firm size in a frictional labor market model. A related paper by Sorkin (2025) uses a model with both statistical and taste-based discrimination to study racial gaps in separations and the implications for racial earnings gaps. My focus is on understanding how racial employment gaps change with aggregate economic conditions, particularly across the firm size distribution.

Finally, the macro literature has studied the role of firm heterogeneity in labor market fluctuations. Empirically, Moscarini & Postel-Vinay (2012) and Haltiwanger *et al.* (2018) show the importance of job creation at large firms for aggregate employment fluctuations. Other papers have introduced firm heterogeneity and endogenous size in the canonical random search model (Elsby & Michaels, 2013, Acemoglu & Hawkins, 2014, Fujita & Nakajima, 2016), and shown that information frictions are important in this context (Baydur, 2017). My paper extends these findings to embed worker group heterogeneity and statistical and taste-based discrimination frameworks.

2. Empirical Evidence

This section studies how employment patterns by race and firm size vary with macroeconomic conditions.

2.1. Data

I use data from the Survey of Income and Program Participation (SIPP), which provides high-frequency information on workers' transitions between employment states and employer types in combination with details about worker occupations, education, and other characteristics (US Census Bureau, 1996). The SIPP is a rotating panel that interviews households every four months for approximately 3-4 years. I use data from the 1996, 2001, 2004, and 2008 panels.⁴ I focus on individuals aged 20 or older who self-identify as non-Hispanic white or Black.

In order to study differences in employment rates and transition rates by employer type, I start by assigning each person to a monthly labor force state using their labor force status for the week corresponding to BLS convention, as described by Fujita *et al.* (2007). I first assign workers as either employed or nonemployed (either unemployed or out of labor force). I focus on nonemployment because I observe a substantial number of transitions from inactivity to employment that I do not want to exclude.⁵ To capture the segment of the population that is reasonably attached to the labor force, I exclude individuals who are inactive over the full SIPP panel (3-4 years). The SIPP has a well-documented issue of seam bias, whereby respondents are more likely to report transitions over the months between survey waves (Moore, 2008). Because one-quarter of respondents are at the seam in each month, this bias should not contaminate the analysis but I perform robustness to dropping observations at the seams.

For employed individuals, I use job and business history information to match employer characteristics to employment status. I assign each employed worker-month observation to

⁴For the 2008 panel, I only use waves 1-10 of 16 due to a change in the firm size survey instrument. See Appendix 7.1 for details on the construction of firm size and the discrepancy in the later waves of the 2008 panel.

⁵Roughly half of all nonemployment-to-employment transitions are from unemployment.

one of four mutually exclusive employer classifications: large firm, small firm, government, or self-employed. Large firms have 100 employees or more, as firm size above this threshold is not further disaggregated during my sample period. For workers who are simultaneously employed by two jobs, I select the job that has higher hours, with longer tenure as a tie-breaker. I only classify a worker as self-employed if they do not work for another employer during that month. I am able to classify 99% of workers to their employer type.⁶

I use job start and end dates to identify hires and separations. I define a hire as a person who meets one of the following criteria: reports a job start date in the current month or was nonemployed in the previous month and employed in the current month. I define a large-firm hire and a small firm hire in the same manner, restricting to large and small firms. Based on these criteria, I observe approximately 167,000 hires, with 51% large-firm hires, 38% small-firm hires.

Similarly, I define a separation as a job with an end date in the current month, or a person employed in the current month but nonemployed in the following month. I classify separations as either voluntary or involuntary based on the reason given for the job ending.⁷ I also classify separations that I do not observe end dates for as involuntary if the worker is unemployed in the following month (not inactive). I observe approximately 155,000 separations, with 51% and 38% at large and small firms. 43% of separations are categorized as involuntary. Appendix 7.1 provides additional summary statistics.

⁶This classification is not 100% because some workers have more than two employers over the four month survey period so I only observe the two that they choose to describe in the interview, or there may be inconsistencies in the start/end dates.

⁷Reasons for involuntary separations include: on layoff, discharged/fired, employer bankrupt, employer sold business, job was temporary and ended, slack work or business conditions.

2.2. Aggregate conditions and relative employment by race and firm size

In this section, I study how employment by race and firm size varies with aggregate conditions in the economy. I start with a linear probability model of employment, with the following specification,

$$E_{ijt} = \alpha_j + \alpha_j^B \text{Black}_i + \beta_j \text{UR}_t + \beta_j^B \text{Black} \times \text{UR}_t + \Gamma_j X_{ijt} + \epsilon_{ijt}, \quad (1)$$

where E_{ijt} is an indicator equal to 1 if worker i is employed at a type- j firm in month t ; Black_i is a racial dummy variable; UR_t is the state-level unemployment rate in month t with units standardized; and X_{it} is a vector of worker characteristics. Worker characteristics are age; age-squared; marital status interacted with gender; education; geographic region; an indicator for large metro area; modal occupation and industry. I cluster standard errors by person and time. I run this regression separately for each employer-type outcome, so the sample is always the same set of workers but the outcome of interest changes. The total employment rate is equal to the sum of the employer-specific rates.

The results are reported in Panel (a) of Table 1. Black workers face lower employment rates relative to white workers with similar characteristics, with the vast majority of the gap arising from small firms, consistent with Figure 1. When the state unemployment rate is 1-standard-deviation higher, the probability of employment decreases across all types of employers. For Black workers, the employment probability decreases by an additional 0.4 percentage point. This additional decrease is driven by large and public employers, though the difference is not statistically significant for large firms.

Given the panel structure of the SIPP, I can analyze how changes in the state unemployment rate affect employment outcomes within individuals using a fixed effects specification,

$$E_{ijt} = \alpha_{ij} + \beta_j \text{UR}_t + \beta_j^B \text{Black} \times \text{UR}_t + \epsilon_{ijt}, \quad (2)$$

Table 1: Employment fluctuations by firm type

(a) Individual controls					
	(1)	(2)	(3)	(4)	(5)
	All	Large	Small	Government	Self
Black	-4.08 (0.22)	2.65 (0.31)	-7.31 (0.25)	2.73 (0.22)	-2.04 (0.17)
UR	-1.20 (0.13)	-0.52 (0.13)	-0.50 (0.11)	-0.06 (0.08)	-0.14 (0.09)
Black $\times$ UR	-0.64 (0.21)	-0.42 (0.28)	0.24 (0.21)	-0.70 (0.19)	0.22 (0.15)
R2	0.069	0.168	0.063	0.387	0.130
N	5,967,568	5,967,568	5,967,568	5,967,568	5,967,568
Black mean	75.87	39.46	14.88	16.32	4.55
White mean	83.88	37.30	21.87	13.20	10.49

(b) Individual fixed effects					
	(1)	(2)	(3)	(4)	(5)
	All	Large	Small	Government	Self
UR	-1.49 (0.27)	-1.30 (0.16)	-0.30 (0.11)	0.03 (0.07)	0.02 (0.06)
Black $\times$ UR	-0.64 (0.29)	-0.62 (0.27)	0.27 (0.25)	-0.34 (0.16)	0.16 (0.12)
R2	0.573	0.739	0.681	0.854	0.829
N	6,097,293	6,097,293	6,097,293	6,097,293	6,097,293
Black mean	75.87	39.46	14.88	16.32	4.55
White mean	83.88	37.30	21.87	13.20	10.49

The table reports differences in employment probability by race and macroeconomic conditions. The units are percentage points. UR is the state-level unemployment rate expressed in standard deviations from the mean. Panel (a) includes controls for age; age-squared; marital status interacted with gender; education; state; metro area size; modal industry and occupation. Panel (b) includes individual fixed effects. Standard errors are clustered by person and month.

where again, the sample is always the same but the outcome variable of interest changes with employer type. I cluster standard errors by person and time. The coefficients on the unemployment rate should pick up differences in the timing of employment within person.

The results are shown in Panel (b) of Table 1. The effect of aggregate conditions on employment probability is slightly larger with individual fixed effects. A 1-standard-deviation increase in the state unemployment rate is associated with a 1.5 percentage point

decrease in the probability of employment for white workers and a 2.1 percentage point decrease for Black workers. The excess reduction in employment is concentrated among large firms, especially for Black workers. Black workers face an additional 0.6 percentage point reduction in employment at large firms. Appendix 7.1 shows that these results are robust to including additional interaction terms.⁸

2.3. Hiring and separation patterns

Next, I consider patterns in hiring and separation by race and firm size. I start with the same linear probability model as Equations 1 and 2, with the outcome variable equal to 1 if person i is a new hire in month t . This specification is distinct from studying job-finding rates, and instead is meant to show how aggregate flows in the total number of hires vary with aggregate conditions, conditional on individual characteristics. It will capture both changes in job-finding rates as well as changes in the composition of employment, since it does not condition on a person’s employment status. The results are shown in Table 2 Panel (a). Within individuals (columns 4-6), hiring is sensitive to the state unemployment rate, with the probability falling 0.14 percentage point for white workers, and an additional 0.23 percentage point for Black workers. This sensitivity of hiring to aggregate conditions is concentrated among large firms, with the hiring probability decreasing 0.14 percentage point, the same as the aggregate effect, and an additional 0.15 percentage point decrease for Black workers.

Table 2 Panel (b) reports variation in job-finding rates, rather than aggregate hiring. Columns (1)-(3) restrict the sample to individuals who were nonemployed in the previous month or who are coded as nonemployed in the current month but report starting a job in this month. Columns (4)-(6) do the same for unemployment. Similar to the results

⁸Online Appendix A.1 reports results separately by gender. Online Appendix A.2 shows that large firms play a more important role in aggregate fluctuations as well, using data from the Quarterly Workforce Indicators.

Table 2: Job inflows by firm type

(a) Hires (percent of population)						
	(1)	(2)	(3)	(4)	(5)	(6)
	All	Large	Small	All	Large	Small
Black	0.04	0.32	-0.31			
	(0.04)	(0.03)	(0.02)			
UR	-0.11	-0.08	-0.05	-0.14	-0.14	-0.02
	(0.03)	(0.02)	(0.01)	(0.10)	(0.05)	(0.03)
Black $\times$ UR	-0.08	-0.10	0.01	-0.23	-0.15	-0.09
	(0.03)	(0.03)	(0.02)	(0.08)	(0.06)	(0.04)
R2	0.011	0.007	0.006	0.075	0.069	0.068
N	5,967,568	5,967,568	5,967,568	6,097,293	6,097,293	6,097,293
Black mean	3.05	1.85	0.90	3.05	1.85	0.90
White mean	2.65	1.32	1.03	2.65	1.32	1.03
Individual FE				X	X	X

(b) Job-finding						
	(1)	(2)	(3)	(4)	(5)	(6)
	All	Large	Small	All	Large	Small
Black	-2.13	0.09	-2.01	-8.07	-2.22	-5.17
	(0.19)	(0.13)	(0.10)	(0.42)	(0.32)	(0.24)
UR	-1.14	-0.64	-0.43	-5.27	-2.86	-2.01
	(0.18)	(0.09)	(0.07)	(0.26)	(0.16)	(0.13)
Black $\times$ UR	-0.18	-0.28	0.11	1.63	0.27	1.25
	(0.15)	(0.11)	(0.08)	(0.30)	(0.23)	(0.16)
R2	0.027	0.021	0.016	0.036	0.026	0.022
N	906,071	906,071	906,071	223,203	223,203	223,203
Black mean	9.68	5.63	2.79	16.69	10.08	4.84
White mean	11.60	5.40	4.32	24.59	12.17	9.41
Unemployed sample				X	X	X

The table reports differences in hiring rates by race and macroeconomic conditions. The units are percentage points. UR is the state-level unemployment rate expressed in standard deviations from the mean. Panel (a) includes all individuals and panel (b) restricts to nonemployed or unemployed individuals. All columns without fixed effects include controls for age; age-squared; marital status interacted with gender; education; state; metro area size; modal industry and occupation. Standard errors are clustered by person and month.

for aggregate hiring rates, job-finding rates decrease when the state unemployment rate is 1-standard deviation higher. Job-finding from nonemployment decreases more for Black workers at large firms, with Black workers experiencing an additional 0.28 percentage point

decrease in the large-firm job-finding probability. In the unemployment specification, the job-finding rate decreases by a much larger 5.3 percentage points for white workers when the state unemployment rate is 1-standard-deviation higher. For Black workers, the decrease in job-finding with increases in state unemployment is smaller, though there is a much larger gap on average. Most of the attenuation comes from small firms, indicating that for Black workers, the gap in job-finding at large firms is relatively more consequential when unemployment is high relative to when it is lower.

Table 3 reports results for separation rates, where I estimate equation 1 on the sample of workers employed in the current month. The outcome variable in columns (1)-(3) is an indicator for any separation between the current month and the next month, and columns (4)-(6) restrict the outcome variable to involuntary separations. The involuntary distinction is important for the aggregate coefficient on the unemployment rate, with columns (1)-(3) having small, negative, and statistically insignificant coefficients, while columns (4)-(6) have larger positive coefficients. Across specifications, the Black-unemployment interaction term is negative and not statistically different from zero. Thus, in the SIPP data, separations for Black workers do not appear to be more sensitive to aggregate unemployment, though the overall Black-white separation gap is large.⁹

3. Model

The empirical results demonstrate that Black employment is more sensitive to aggregate conditions than white employment, particularly at large firms, in excess of what would be predicted by worker characteristics. The pattern holds for both employment and hiring outcomes. In this section, I develop a model that embeds heterogeneous advantages in the hiring process as a mechanism for contributing to these patterns. The model features

⁹Separations for Black men are weakly more responsive to aggregate unemployment, as shown in Online Appendix A.1.

Table 3: Separation rates by firm type

	(1)	(2)	(3)	(4)	(5)	(6)
	All	Large	Small	All	Large	Small
Black	0.26 (0.05)	0.40 (0.06)	0.51 (0.11)	0.35 (0.03)	0.45 (0.04)	0.52 (0.07)
UR	-0.03 (0.04)	-0.05 (0.04)	0.00 (0.05)	0.18 (0.02)	0.18 (0.03)	0.29 (0.03)
Black $\times$ UR	-0.03 (0.04)	-0.10 (0.06)	0.03 (0.10)	-0.01 (0.03)	-0.00 (0.04)	-0.04 (0.08)
R2	0.012	0.013	0.010	0.006	0.006	0.006
N	4,826,652	2,291,178	1,283,686	5,050,241	2,291,178	1,283,686
Black mean	6.61	6.66	10.58	3.07	2.84	4.76
White mean	3.17	3.05	4.08	1.27	1.23	1.79
Involuntary				X	X	X

The table reports differences in separation rates by race and macroeconomic conditions. The units are percentage points. Columns (1)-(3) include all separations and columns (4)-(6) restrict the outcome variable to involuntary separations. UR is the state-level unemployment rate expressed in standard deviations from the mean. All columns include controls for age; age-squared; marital status interacted with gender; education; state; metro area size; modal industry and occupation. Standard errors are clustered by person and month.

heterogeneous firms, heterogeneous workers, and a frictional labor market. It is set in discrete time.

3.1. Environment

Workers A unit mass of infinitely-lived workers are endowed with one indivisible unit of labor. They share a common discount factor, β , with linear preferences for consumption. They produce and consume a single homogeneous good. Workers have no disutility of labor but may be unemployed due to frictions in the labor market. Let u_t denote the mass of unemployed workers at the start of period t (with $1 - u_t$ the mass of employed workers). Unemployed workers receive flow utility b . Firms are owned by workers with dividends distributed in lump sum.

There are two types of workers, $g \in \{W, B\}$, corresponding to white and Black, with (a fixed) fraction $\pi < \frac{1}{2}$ in B . Workers are ex-ante identical. I give three possible channels for

racial disparities in the model: (1) firms receive noisier information about Black candidates, as in the statistical discrimination literature; (2) firms have a disutility for employing Black workers, as in the taste-based discrimination literature; (3) the racial composition of matches is skewed towards the racial composition of incumbent employment, to provide an avenue for coracial hiring.

Firms There are two types of firms indexed by their idiosyncratic productivity z , which I assume to be fixed. There is no firm entry or exit in this model.¹⁰ Let $\nu(z)$ be the share of firms with productivity z . They share a common aggregate productivity a . They use labor to produce a single good with decreasing returns to scale production technology,

$$y = azn^\alpha.$$

Differences in productivity with decreasing returns to scale generates large and small firms. Firms also differ in their recruiting costs, described below.

Matching and hiring process This is a random search model with information frictions in the hiring process (Baydur, 2017). Firms post vacancies (v) to attract matches at cost $c_v(z)$, which varies with firm productivity. Vacancy posting is an intensive-margin decision. The more vacancies the firm posts, the more candidates it has to choose from when deciding who to hire. The matching rate between vacancies and unemployed workers depends on market tightness, θ , where the probability that a vacancy attracts a worker is $q(\theta)$, the probability an unemployed worker meets a firm is $\theta q(\theta)$, and $\theta = \frac{V}{U}$ is market tightness.

Matches are allocated via random search with a distortion towards the racial composition of employment, meant to reflect that firms that have a higher share of Black employment

¹⁰I address this assumption in the quantitative section for the interpretation of the mapping between the data and the model. Appendix 7.1 provides additional empirical results that exclude firm exit.

may attract more Black matches, for example due to differences in network effects as in [Miller & Schmutte \(2023\)](#). Without this channel, firms would sample workers proportional to their share in the unemployed pool and workers would sample firms proportional to their share of the total vacancies. For each firm, I allow the total number of matches to be the same as the undistorted case, i.e. each firm has probability $q(\theta)$ of meeting a worker, but I allow the composition of matches to skew towards the racial composition of employment.¹¹ I define the number of matches between a worker from group g and a firm of productivity z to be $q(\theta)v_z\lambda_{gz}$, with

$$\lambda_{gz} \equiv \lambda \frac{n_{gz}}{n_z} + (1 - \lambda) \frac{u_g}{u}$$

The parameter λ governs the degree of distortion in matching. I assume that firms take λ_{gz} as given and do not internalize the effect of hiring an additional group g worker on future recruitment.

When workers and firms meet, both parties face uncertainty around the worker's productivity, which is revealed at the production stage if the worker is hired. Workers can either be a productive type, contributing one unit of labor to the firm's production function, or unproductive, contributing zero. Each time a worker meets a firm, they draw a new match quality from the same distribution, $F(\cdot)$, which determines the likelihood the worker will be productive.¹² Match quality is unobservable to the worker and the firm, but both observe a signal of match quality, s . The signal follows the inspection technology form of [Menzio & Shi \(2011\)](#), where the firm observes the true match quality with probability p . With probability

¹¹This creates a disadvantage for the under-represented group because matches will be lower in absolute terms.

¹²This set-up follows [Baydur \(2017\)](#) to create a learning model similar to [Jovanovic \(1979\)](#) in which all uncertainty is resolved within the first period. This creates a distribution of types in the hiring stage without adding the distribution of types to the state space.

$1 - p$, the firm observes another i.i.d. draw from the same distribution. Thus, the firm forms a posterior belief (x) about the worker’s productivity conditional on their signal, according to

$$x = ps + (1 - p)\mathbb{E}[s]. \quad (3)$$

I allow this probability to vary by race and firm type, p_{gz} . This allows for a statistical discrimination channel in the model, where firms may receive noisier information about one group, leading them to place more weight on the population mean than the individual’s qualifications (e.g. [Black, 1995](#), [Lang & Lehmann, 2012](#)). Differences in noise could also capture differences in referral networks if these affect the information firms have about potential hires, as in [Miller & Schmutte \(2023\)](#).

Using these beliefs, the firm must decide which matches to hire. The firm chooses a group-specific threshold rule, x_g^* such that it hires all matches from that group with expected productivity above the threshold. Once workers are hired, wages are bargained using [Stole & Zwiebel \(1996\)](#) and then wages are paid, production occurs, and new hire types are revealed. The final racial disparity I allow for in the model is taste-based discrimination, where firms have utility costs over groups of workers, χ_{gz} . I assume these are paid for all incumbent workers and new hires at the production stage. I anticipate stronger taste costs for Black workers at small firms, which could arise if small firms have stronger preferences for hiring workers of the same race, given the higher prevalence of white-owned businesses. It could also reflect the lower scrutiny that small employers face from the Equal Employment Opportunity Commission (EEOC), perhaps allowing more scope for taste-based discrimination.¹³

At the start of the next period, all of the unproductive hires from the end of the previous period separate and an exogenous share δ of the productive hires separate. These

¹³One interpretation could be that the taste costs are net of penalties, which are likely lower for small firms, as the EEOC tends to prioritize its limited resources for large firms.

newly separated workers are not able to search until the following period.

3.2. Optimization

Firm's Problem The firm chooses vacancies, v , and hiring standards, $\{x_g^*\}$, which implicitly define the number of hires, $\{h_g\}$, the expected productivity of hires, and next period employment, $\{n'_g\}$ for each group. The idiosyncratic states are the number of incumbent workers from each group and fixed firm productivity, z . The aggregate state space is rich. It includes aggregate productivity, a , market tightness, θ , and other distributional features of the economy, such as the Black share of unemployed workers. To ease notation, I use the subscript t to summarize the aggregate states in the firm's value function,

$$J_t(n_B, n_W, z) = \max_{v \geq 0, x_g^*} -c_v(z)v + a_t z (n')^\alpha - \sum_g ((1-\delta)n_g + h_g) \chi_{gz} \quad (4)$$

$$- \sum_g ((1-\delta)n_g w_t^n(n', z, g) + h_g w_t^h(x_g^*, n', z, g)) + \beta \mathbb{E}_t J_{t+1}(n'_B, n'_W, z)$$

such that

$$n' = \sum_g n'_g \quad (5)$$

$$n'_g = (1-\delta)n_g + \lambda_{gz} q(\theta_t) v \int_{x \geq x_g^*} x dF(x|p_{gz}) \quad (6)$$

$$h_g = \lambda_{gz} q(\theta_t) v (1 - F(x_g^*|p_{gz})) \quad (7)$$

$$(1 - p_{gz})\mathbb{E}[x] \leq x_g^* \leq p_{gz} + (1 - p_{gz})\mathbb{E}[x] \quad (8)$$

where $w_t^n(\cdot)$ and $w_t^h(\cdot)$ are wages paid to incumbent workers and new hires, respectively. These wages are determined by bargaining and may depend on aggregate and idiosyncratic states.¹⁴ Equation 8 reflects the bounds of posterior belief formation from equation 3 with the signal

¹⁴With a slight abuse of notation, the new-hire wage here is the average wage.

ranging from 0 to 1. $F(x|p)$ is the cumulative distribution of posteriors conditional on signal quality p ,

$$F(x|p) = F\left(\frac{x - (1-p)\mathbb{E}[x]}{p}\right), \quad (9)$$

where $F(\cdot)$ is the exogenous distribution of match quality.

Vacancy costs are linear but I allow the vacancy cost to vary with fixed firm productivity, z , with the assumption that $\frac{\partial c_v(z)}{\partial z} < 0$. Thus firms with higher productivity, which will be endogenously larger, have lower vacancy costs. In a two-firm model, this specification delivers the intuition that larger firms can have larger human resources departments or other economies of scale that lets them screen applicants at a lower marginal cost without introducing additional size dependence or time inconsistency in the bargaining problem with workers.

The firm's problem summarizes the three sources of racial disparities in the model. The number of hires depends on the distortion in matching rates, λ_{gz} . The threshold rules depend on the quality of the signal, p_{gz} . The composition of employment affects total taste costs, χ_{gz} . The three channels interact through the firm's problem to determine the equilibrium disparities in employment overall.

Worker's Problem Let $V_t^u(g)$ be value of unemployment for a worker from group g at the end of the period and $V_t^n(n, g, z)$ be the value of a worker employed at a firm of type z that is known to be productive,

$$V_t^n(n, g, z) = w_t^n(n, z, g) + \beta \mathbb{E}_t [V_{t+1}^u(g) + (1 - \delta)(V_{t+1}^n(n', g, z) - V_{t+1}^u(g))], \quad (10)$$

where n is the number of productive workers. Newly hired workers can be paid different wages and face higher separation rates, captured in the value function,

$$V_t^h(x, n, g, z) = w_t^h(x, n, z, g) + \beta \mathbb{E}_t [V_{t+1}^u(g) + x(1 - \delta)(V_{t+1}^n(n', g, z) - V_{t+1}^u(g))], \quad (11)$$

where x is the probability the worker is productive, conditional on their signal. For unemployed workers, the value function is

$$V_t^u(g) = b + \beta \mathbb{E}_t V_{t+1}^u(g) \quad (12)$$

$$+ \underbrace{\beta \mathbb{E}_t \left[\theta_{t+1} q(\theta_{t+1}) \sum_z \tilde{\lambda}_{gz} \int_{x > x_{gz}^*} (V_{t+1}^h(x, n', g, z) - V_{t+1}^u(g)) dF(x|p_{gz}) \right]}_{\Omega_t(g)} \quad (13)$$

$$\tilde{\lambda}_{gz} = \lambda_{gz} \left(\frac{\mu_z v_z}{\sum_{\tilde{z}} \mu_{\tilde{z}} v_{\tilde{z}}} \right) \left(\frac{u_g}{u} \right)^{-1}$$

where v_z is the equilibrium number of vacancies posted by a firm of type z , μ_z is the (exogenous) mass of type z firms per worker in the economy, x_{gz}^* is the firm's equilibrium threshold rule, and $\tilde{\lambda}_{gz}$ is the matching distortion from the perspective of the worker. If there is no incumbent bias in matching, $\lambda = 0$, then $\lambda_{gz} = u_g/u$ and $\tilde{\lambda}_{gz}$ is equal to the vacancy share for type z firms. If $\lambda > 0$, then $\sum_z \tilde{\lambda}_{gz}$ may be higher or lower than 1 depending on whether group g is over- or under-represented in employment.

Wage bargaining Wages are set via [Stole & Zwiebel \(1996\)](#) bargaining in which firms bargain with each worker sequentially and failure to negotiate with a worker requires them to go back and bargain again with the others.¹⁵ Firms and workers split the surplus according

¹⁵This is a standard bargaining rule in models with endogenous firm size, such as [Acemoglu & Hawkins \(2014\)](#), [Baydur \(2017\)](#), and [Elsby & Michaels \(2013\)](#).

to the following rules

$$\phi D_{t,n_g} = (1 - \phi) (V_t^n(n', g, z) - V_t^u(g)) \quad (14)$$

$$\phi D_{t,x,g} = (1 - \phi) (V_t^h(x, n', g, z) - V_t^u(g)) , \quad (15)$$

where ϕ is the worker's bargaining power, D_{t,n_g} is the marginal surplus to the firm of having one more incumbent worker from group g , and $D_{t,x,g}$ is the marginal surplus to the firm of having one more hire from group g with probability x of being productive, both measured after vacancy costs are sunk and hiring thresholds have been chosen. The full details are provided in Appendix 7.2.

Using the firm and worker value functions with the sharing rules, I get the following equilibrium wage functions,

$$w^n(n', z, g) = \frac{\alpha\phi}{1 - \phi + \alpha\phi} a_t z n'^{\alpha-1} + (1 - \phi)(b + \Omega_t(g)) - \phi\chi_{gz} \quad (16)$$

$$w^h(x, n', z, g) = x \frac{\alpha\phi}{1 - \phi + \alpha\phi} a_t z n'^{\alpha-1} + (1 - \phi)(b + \Omega_t(g)) - \phi\chi_{gz}, \quad (17)$$

where $\Omega_t(g)$ is the value of searching next period for a worker from group g as defined in equation (12). This term is included in addition to the flow value of unemployment, b , because workers who separate are not able to search in the following period.

Notice that the first component of the wage equation is a fraction of the marginal product of labor. The second component depends on workers' outside options. If groups face different prospects in the labor market, the second term will vary by group, and thus within-firm wage dispersion may arise.¹⁶ The third component represents the pass-through of taste-based discrimination.

¹⁶This could be microfounded by a job-ladder model where wages depend on outside offers. If workers have different professional networks and barriers to receiving offers, then wages would differ as a result of these frictions.

Aggregation Let μ_z be the mass of type z firms (relative to a unit mass of workers). The mass of unemployed workers in each group evolves according to

$$u_{gt+1} = \pi(g) - \sum_z \mu_z \left(n'_{gz} + \lambda_{gz} q(\theta_t) v_z \int_{x \geq x_g^*} (1-x) dF(x|p) \right), \quad (18)$$

where $\pi(g)$ is the share of group g in the population, n'_{gz} is the number of incumbent workers at the start of the next period, and the second term in the sum represents the number of hires who will separate in the next period because they are revealed to be unproductive. These workers are not able to search in the following period and should be excluded from the unemployment rate.

The distribution of employment across firms is given by

$$\Gamma(z) = \frac{\mu_z \sum_g ((1-\delta)n_{gz} + h_{gz})}{\sum_{\bar{z}} \mu_{\bar{z}} \sum_g ((1-\delta)n_{g\bar{z}} + h_{g\bar{z}})} \quad (19)$$

3.3. Equilibrium

Equilibrium definition Given exogenous masses of firms μ_z , a recursive competitive equilibrium for this economy is a list of functions: (i) value functions for firms, $J_t(n_B, n_W, z)$, (ii) decision rules for vacancies and hiring standards, $v_t(n_B, n_W, z)$, $x_{tg}^*(n_B, n_W, z)$, (iii) value functions for workers $V_t^n(n', g, z)$, $V_t^h(x, n', g, z)$, $V_t^u(g)$, (iv) wage functions $w_t^n(n', g, z)$, $w_t^h(x, n', g, z)$, and (v) worker outside option functions $\Omega_t(g)$; and market tightness θ , a stationary distribution of employment across firms, $\Gamma(z)$, and a stationary distribution of each group of workers in unemployment and each employer type, $\{u_g, n_{gz}\}$.

1. *Firm optimization*: Given θ , $\{u_g, n_{gz}\}$, $\Omega_t(g)$, $w_t^n(n', z, g)$, and $w_t^h(x, n', z, g)$, the set of decision rules $v_t(n_B, n_W, z)$, $x_{tg}^*(n_B, n_W, z)$ solve the firm problem;
2. *Worker optimization*: Given θ , $\Gamma(z)$, $w_t^n(n', z, g)$, $w_t^h(x, n', z, g)$, and $v_t(n_B, n_W, z)$, $x_{tg}^*(n_B, n_W, z)$, worker value functions $V_t^n(n', g, z)$, $V_t^h(x, n', g, z)$, and $V_t^u(g)$ solve the worker problem

and $\Omega_t(g)$ is consistent with value functions;

3. *Wage bargaining*: $w_t^n(n', z, g)$, $w_t^h(x, n', z, g)$ solve the bargaining problem;
4. *Consistency*: The stationary distribution of employment $\Gamma(z)$ is consistent with firm optimization;
5. *Market clearing*: The labor market clears and the distribution of workers across un-employment and employer types, $\{u_g, n_{gz}\}$, is consistent with firm optimization.

Firm's problem solution With the wage equations, the firm's problem can be rewritten as choosing the number of productive workers from each group, subject to a cost minimization problem,

$$\begin{aligned}
J_t(n_B, n_W, z) = & \max_{n'_g \geq (1-\delta)n_g} -C_t(\Delta_B, \Delta_W) + \frac{1-\phi}{1-\phi+\alpha\phi} a_t z (n'_B + n'_W)^\alpha \\
& - \sum_g n'_g \left((1-\phi)(b + \Omega_t(g) + \chi_{gz}) \right) + \beta \mathbb{E}_t J_{t+1}(n'_B, n'_W, z) \\
\text{s.t.} \\
\Delta_g = & n'_g - (1-\delta)n_g,
\end{aligned}$$

where

$$\begin{aligned}
C_t(\Delta_B, \Delta_W) = & \min_{\{x_g^*\}} \sum_g \Delta_g \frac{c_v(z)}{q(\theta_t) \int_{x \geq x_g^*} x dF(x|p_{gz})} \\
& + \Delta_g \frac{\int_{x \geq x_g^*} (1-x) dF(x|p)}{\int_{x \geq x_g^*} x dF(x|p_{gz})} (1-\phi)(b + \Omega_t(g) + \chi_{gz}) \quad (20)
\end{aligned}$$

s.t.

$$\frac{\Delta_B}{\lambda_{Bz} \int_{x \geq x_B^*} x dF(x|p_{gz})} = \frac{\Delta_W}{\lambda_{Wz} \int_{x \geq x_W^*} x dF(x|p_{gz})} \quad (21)$$

and $C_t(\Delta_B, \Delta_W)$ can be understood as the total cost of hiring $\Delta_B + \Delta_W$ *productive* workers. The first term inside the cost minimization sum is the total expenditure on vacancies that match to group g workers. The second term is the total expenditure on outside option compensation and taste costs for workers who are not productive. Consider the limit as x_g approaches the upper bound. The first term will go to infinity as the number of vacancies needed to get Δ_g productive workers goes to infinity. The second term will go to zero. As x_g approaches the lower bound, the first term goes to $\frac{c_v(z)}{q(\theta)\bar{x}}$, which is the minimum possible expenditure on vacancies. The second term goes to $\frac{1-\bar{x}}{\bar{x}}(1-\phi)(b + \Omega_t(g) + \chi_{gz})$, which is the maximum possible expenditure on unproductive compensation and taste costs, reflecting unconditional probabilities. The cost-minimizing threshold, x_g^* lies somewhere between the two extremes.

Finally, the constraint (21) reflects the implications of random search: firms cannot post separate vacancies for different groups of workers. Hiring an additional productive worker from one group increases the vacancy posting expenditure for the other group, since some of the additional vacancies will match to other workers.

For an interior solution, the firm will add productive hires until the shadow value of an additional group g worker, Λ_{gzt} , is equal to the marginal cost of hiring another:

$$\Lambda_{gzt} = \frac{\partial C_t(\Delta_B, \Delta_W)}{\partial \Delta_g} \quad (22)$$

$$\Lambda_{gzt} = \frac{\alpha(1-\phi)}{1-\phi+\alpha\phi} a_t z (n')^{\alpha-1} - (1-\phi)(b + \Omega_{t+1}(g) + \chi_{gz}) + \beta(1-\delta)\mathbb{E}_t[\Lambda_{gzt+1}], \quad (23)$$

where the shadow value of a type g worker is equal to the marginal product of labor net of wages plus the expected shadow value of that worker in the next period.

The cost minimization solution replicates a free-entry condition for the intensive margin

of vacancies,

$$\frac{c_v(z)}{q(\theta_t)} = \sum_g \lambda_{gz}(1 - \phi)(b + \Omega_t(g) + \chi_{gz}) \frac{1}{x_g^*} \left(\int_{x \geq x_g^*} x dF(x|p_{gz}) - x_g^*(1 - F(x_g^*|p_{gz})) \right). \quad (24)$$

The left side is the marginal vacancy cost, times the expected duration of the vacancy. The right side reflects the marginal vacancy benefit. This depends on the probability that the firm hires the match, $x \geq x_g^*$, and conditional on hiring the worker, whether the worker is productive. With probability x , the worker is productive and the firm gets the shadow value of an additional productive worker from group g , Λ_{gz} . With probability $(1 - x)$, the worker is not productive and the firm pays additional compensation and taste costs, $(1 - \phi)(b + \Omega_t(g) + \chi_{gz})$. This marginal benefit simplifies to the expression shown, which does not explicitly include the shadow value of a productive worker. This expression is monotonically decreasing in x_g^* . This can be understood as the trade-off between hiring strategies—firms can either post more vacancies or become less selective about the matches they accept. Intuitively, when the cost of getting an additional match is relatively lower (left-hand side), then firms will be more selective (high x^*). With only one group, this equation would pin down the hiring threshold independently of productive hires, Δ , as in [Baydur \(2017\)](#). With two groups, the threshold decisions are jointly determined by equations (21) and (24), and the threshold rule depends on relative productive hires between the two groups due to spillovers from random search.

4. Calibration

I calibrate the model at a monthly frequency. I first fix a set of parameters using moments from the data or external estimates. Then, I calibrate the remaining parameters to match moments from the data using generalized method of moments.

4.1. Empirical moments

I use data from the SIPP for the empirical moments in the model using the sample described in Section 2.1. I take the unconditional share of Black workers in the sample as given. The other relevant moments include the employment, job-finding, and separation rates by firm size and race. I start with the average employment rate for all workers in my sample. Then, I use the coefficient on the Black dummy variable in Table 1 Panel (a) and the unconditional share of Black workers in the sample to back out employment rates for Black and white workers.¹⁷ I repeat this process for job-finding and separation rates using the Black shares of employed or nonemployed populations and gap estimates from Tables 2 and 3. I then compute size-specific moments for large- and small-firm employment, job-finding and separation rates by rescaling the gap estimates from each table such that the large- and small-firm coefficients sum to the aggregate. For example, the aggregate racial employment gap in Table 1 is -4.1 percentage points and I define the large-firm gap to be 2.3 percentage point using this methodology.¹⁸

4.2. Fixed parameters

The top panel of Table 4 summarizes the values of fixed parameters and their sources. Given the monthly frequency, I set the discount factor β to 0.996 to match a quarterly interest rate of 0.012. I set the production curvature α to 0.677, as is common in the RBC literature. I use a standard Cobb-Douglas matching technology, $q(\theta) = \zeta\theta^{-\psi}$, with matching elasticity $\psi = 0.6$, as in Petrongolo & Pissarides (2001).

For the exogenous distribution of match quality, I use the functional form assumption

¹⁷With the Black share of the population of 13.6% and the aggregate employment rate equal to 82.79%, this gives employment rates of 79.3% and 83.4% for Black and white workers, respectively. The employment gap is 4.1 percentage points as in the first row of Table 1 column (1).

¹⁸Using the coefficient estimates from the first row of Table 1, $2.32 = \frac{2.65}{2.65-7.31} \times -4.08$.

from [Baydur \(2017\)](#),

$$F(y) = (y)^{1/(\gamma-1)},$$

with $\gamma > 1$ and $y \in [0, 1]$. This distribution is convenient because it is governed by a single parameter. The unconditional mean of match quality is $1/\gamma$. Higher values of γ will imply that screening is more valuable because the ex-ante quality of the pool is lower. I set $\gamma = 3$, following their calibration. I choose a normalization for signal quality in the baseline model of $p = 0.99$. I use the same normalization across large and small firms, such that statistical discrimination arises from the gap in signal qualities between workers across groups within the same firm.

The masses of high and low productivity firms are fixed in the model, but employment at each type of firm adjusts endogenously. Given the mass of firms calibrated below, I set the share of high-productivity firms to 2%. This parameter comes from the share of firms with 100 or more employees, excluding firms with zero employment, from 1997 Census data, as reported by [Axtell \(2001\)](#). This is the same threshold for defining large firms that I use in the SIPP, and the time period is consistent with my sample that starts in 1996. Given the endogenous relative employment shares, aggregate productivity a scales the absolute value of firm size up, and I choose a value of 4.2, which corresponds to small firms having about 30 employees in equilibrium and large firms having 2,800. The Black share of the population is fixed at 0.136 based on the share of Black relative to white population in the SIPP sample described in Section [2.1](#).

4.3. Fitted parameters

The calibrated parameters affect all moments, but I give a sense of the identification for each. The parameter values are summarized in the bottom panel of Table [4](#).

The overall job-finding rate in the SIPP is the matching rate from the perspective of

Table 4: Model Parameters

Parameter	Meaning	Value	Target
<i>Fixed Parameters</i>			
β	Discount factor	0.996	Quarterly interest rate 0.012
α	Production curvature	0.677	RBC literature
γ	Match quality shape	3	Baydur (2017)
p	Signal quality	0.99	Normalization
v	Share of high-productivity firms	0.02	Axtell (2001)
a	Aggregate productivity	4.2	Relative sizes
π	Black share population	0.136	SIPP
ψ	Matching elasticity	0.6	Petrongolo & Pissarides (2001)
<i>Fitted Parameters</i>			
ζ	Matching scale	0.640	Hired share of matches 0.2
μ	Number firms/worker	0.009	Market tightness 0.72
b	Flow value unemp	0.980	Normalize outside option
ϕ	Bargaining power	0.353	Ratio b to avg prod
δ	Exog. separation	0.024	Average separation rate
$c_v(L)$	Vacancy cost	0.013	Large-firm job-finding rate
$c_v(S)$	Vacancy cost	0.181	Small-firm job-finding rate
$\frac{z_L}{z_S}$	Relative productivity	4.080	Relative large-firm employment
λ	Matching distortion	0.635	Large-firm contact gap, Kline et al. (2022)
Δp_L	Signal gap, large	0.191	Separation gap, large
Δp_S	Signal gap, small	0.284	Separation gap, small
$\Delta \chi_L$	Taste penalty, large	0.002	Sorting gap
$\Delta \chi_S$	Taste penalty, small	0.010	Employment gap

the worker, $\theta q(\theta)$ times the vacancy-weighted average hiring rate across firms and worker groups,

$$\zeta \theta^{1-\psi} \sum_z \sum_g \lambda_{gz} \frac{v(z)}{v} (1 - F(x_{gz}^* | p_{gz})).$$

Thus, given a target for the job-finding rate, ζ governs how selective the firm is. If ζ is low, then the share of matches that are hired increases, whereas if ζ is high, this share decreases. I target 20% as the weighted average of the hired share of matches, which is consistent with firms interviewing around five applicants before making an offer, as in [Barron et al. \(1997\)](#)

or Villena-Roldan (2012).

Aggregation across firms means that market tightness is given by

$$\theta = \frac{\mu(vv_L + (1-v)v_S)}{u_B + u_W},$$

thus the exogenous mass of firms is important for scaling market tightness. I target a market tightness of 0.72 as in Elsby & Michaels (2013).

Following the strategy of Baydur (2017), I use a normalization for b such that the equilibrium value of unemployment for white workers ($b + \Omega(W)$) is equal to 1 and I discipline ϕ by targeting 0.73 for the ratio of the flow value of unemployment, b , to average productivity (Y/N). To see the intuition, consider the shadow value of a white worker in equation 23 and suppose ϕ rises. All else equal, the second term will become smaller in absolute value, so for the shadow value to remain constant, the first term must also fall. This means employment will rise, leading to a decrease in average productivity and an increase in the ratio of b to Y/N .

The exogenous separation rate δ is identified by the average separation rate. The vacancy costs by firm size are identified by the job-finding rates by firm size. As the vacancy cost increases, the firm posts fewer vacancies and job-finding falls. The relative productivity of large firms, $\frac{z_L}{z_S}$ is identified by the employment share at large firms. If the model had no heterogeneity other than differences in firm productivity, large firms would make the same decisions as small firms but with more workers, because z_L would lead them to hire until their marginal product of labor was the same.

I now turn to the parameters that govern disparities between racial groups. I calibrate the matching distortion towards incumbent employment, λ to target the contact gap at large firms documented by Kline *et al.* (2022). In the model, I define the contact gap by firm type to be the gap in matching rates for white workers relative to Black workers at each type of

firm,

$$\frac{\lambda_{Wz}/u_W}{\lambda_{Bz}/u_B} - 1 = \frac{\lambda \frac{n_{Wz}}{n_z} \left(\frac{u_W}{u}\right)^{-1} + 1 - \lambda}{\lambda \frac{n_{Bz}}{n_z} \left(\frac{u_B}{u}\right)^{-1} + 1 - \lambda} - 1$$

Without any other differences between groups, λ will not generate a contact gap because $\frac{n_{gz}}{n_z} \left(\frac{u_g}{u}\right)^{-1} = 1$. Similarly, there will be no contact gap if $\frac{n_{gz}}{n_z} \left(\frac{u_g}{u}\right)^{-1} \neq 1$ unless $\lambda > 0$. In partial equilibrium, the contact gap is increasing in λ . In general equilibrium, higher λ affects the composition of incumbent employment relative to unemployment as well.

Both statistical discrimination and taste-based discrimination would lead to worse average employment for Black workers and more sorting towards large firms if the frictions are stronger at small firms. I separately identify the channels by targeting separation gaps as well. For statistical discrimination, I normalize the signal quality for white workers to be $p_{Wz} = 0.99$, such that what matters is the gap in signal quality between Black and white workers, $\Delta_z = p_{Wz} - p_{Bz}$. For taste-based discrimination, I normalize $\chi_{Wz} = 0$, such that χ_{Bz} can be interpreted as the relative taste costs of employing a Black worker over a white worker.

To understand the identification of each channel, first consider a model with just taste-based discrimination. Stronger taste frictions would lead firms to be more selective about hiring Black workers because of the direct costs of employing them. The direct effect would be unambiguously lower job-finding and employment rates and also lower separation rates for Black workers, as the higher selectivity means the marginal Black applicant is more positively selected and therefore less likely to separate. The statistical discrimination parameters are important for generating positive separation gaps. Lower information quality about Black applicants means that even if firms set the same cutoff x^* across both groups, more Black workers will be revealed to be unproductive because the signal is noisier.¹⁹ Without differ-

¹⁹This is the same result from [Morgan & Várdy \(2009\)](#).

ences in signal quality, if a firm is more selective about hiring it necessarily implies lower separation rates for the affected group. This friction breaks that relationship, which allows the model to match both negative job-finding gaps and positive separation gaps.

Table 5: Moments

(a) Aggregate moments			(b) Racial gap moments		
Moment	Data	Model	Moment	Data	Model
Employment share			Employment, ppt (B-W)		
Large	0.64	0.64	Empl rate	-4.08	-4.08
			Large-firm share	6.20	6.19
Job-finding rate, ppt			Job-finding, ppt (B-W)		
Large	6.45	6.45	Large	0.10	0.00
Small	4.78	4.78	Small	-2.23	-1.23
Separation rate, ppt	3.17	3.17	Separation, ppt (B-W)		
Large	3.05	2.84	Large	0.40	0.40
Small	4.07	3.78	Small	0.51	0.51
External			Contact gap	0.36	0.13
Hired share matches	0.20	0.20	Large firms	0.09	0.09
Market tightness	0.72	0.72			
Outside option	1.00	1.00			
Ratio b to avg prod	0.73	0.73			

The figures in bold represent untargeted moments. The data moments are all calculated in the SIPP except for the external moments and contact gaps.

The model fits the targeted moments closely, as shown in Table 5. I match the average separation rate by construction, but the model matches the distribution across firm size reasonably well. To benchmark how well the contact gap in the model overall matches the data, I compare the size of the aggregate gap to the average 36% gap in the correspondence study literature, as surveyed by [Quillian *et al.* \(2017\)](#). As a more conservative approach, I report the aggregate gap in the model as the weighted average of small and large firms using vacancy shares as weights. This leads the aggregate number to skew closer to the gap for large firms. If I compute the aggregate gap using the relative masses of firms instead, the model counterpart is 28%.

The racial gaps in the model are determined by the interaction of the bias in vacancy

Table 6: Racial employment gaps and sources of disparities

	Baseline	Matching distortion removed	Taste discrimination removed	Statistical discrimination removed
Employment rate	-4.08	-2.69	-4.13	0.79
Large-firm share	6.20	0.76	1.87	5.29

The table reports the gap in percentage points between the Black and white employment-population ratios and the share of employment at large firms. Baseline shows the main calibration. Matching distortion column removes the incumbency bias in vacancy-matching rates ($\lambda = 0$). Statistical discrimination removes additional signal noise for Black workers ($\Delta p_L = \Delta p_S = 0$). Taste discrimination removes taste bias ($\Delta \chi_L = \Delta \chi_S = 0$).

matching towards the composition of incumbent employment and the two forms of discrimination. Table 6 shows what happens to each of these gaps when one of the three elements is removed.²⁰ Removing the matching distortion almost entirely removes the sorting towards large firms and improves the overall employment gap. In the baseline model, Black workers represent 15.8% of unemployment but capture 14.3% of vacancies. Removing incumbency bias results in more Black matches, leading to more Black employment, particularly at small firms. Removing taste-based discrimination leads to less sorting but does not materially change the employment gap. Taste costs create spillovers between small and large firms to the extent that they pass through to wages. Worse taste costs at small firms increase the amount of reallocation of Black workers towards large firms by dampening workers' outside options. Meanwhile, removing statistical discrimination reverses the employment gap but does not meaningfully change the amount of sorting. Without statistical discrimination, small firms are much more selective about hiring Black workers due to taste-based discrimination. This lowers equilibrium wages, which leads large firms to be relatively less selective about hiring Black workers. Given the same information about candidates, this decrease in selectivity leads to higher Black employment at large firms, and is reinforced by the matching distortion.

²⁰Appendix 7.3 shows additional exercises that recalibrate the model after removing each of the channels.

5. Counterfactuals

5.1. Aggregate conditions and racial employment gaps

I examine the relationship between aggregate labor market conditions and racial employment gaps by comparing the baseline steady state equilibrium to a counterfactual steady state with exogenously lower aggregate productivity and endogenously lower market tightness and higher unemployment. I compare the counterfactual change in the racial employment gap across steady states to the empirical patterns reported in Table 1, which show the excess response of Black employment to higher aggregate unemployment. To render the scale of these exercises comparable, I calibrate the decrease in aggregate productivity, a , such that the employment rate for white workers decreases by 1.49 percentage point, reflecting the average decrease in employment for white workers when the state-level unemployment rate is 1-standard-deviation higher.²¹

Table 7 reports the results of this exercise. By construction, the data and model match exactly in the first row for the total change in employment for white workers. Since the data include additional types of employment that are not modeled, I show two versions of the data moments that aggregate large and small contributions either by redistributing the aggregate or changing the aggregate.²² The next two rows compare the decreases in employment at each type of firm between the data and the model. The model is relatively consistent with the data in terms of the shares attributed to each type of firm, though the decrease in employment is less skewed towards large firms than it is in the data.

The second group of Table 7 shows the difference in the racial employment gap be-

²¹This results in a 0.011 log point decrease in productivity.

²²Another difference between data and model is that the data may reflect firm entry/exit that is outside the scope of the model. Appendix Table 10 shows that the regression results are very similar using a sample of workers who never report losing their job due to firm closure or sale, which excludes the exit margin.

Table 7: Steady state comparison

	Data reweight	Data sum	Model
Δ White employment rate	-1.49	-1.60	-1.49
Large	-1.20	-1.30	-1.04
Small	-0.27	-0.30	-0.45
Δ Employment gap	-0.61	-0.35	-1.21
Large	-1.15	-0.62	-1.21
Small	0.54	0.27	0.01

The table compares the change in moments when aggregate labor market conditions worsen between the data and the model. For the data, the change in white employment comes from the regression coefficient on the state-level unemployment rate and the change in the employment gap comes from the coefficient on the Black dummy variable interacted with the state-level unemployment rate in Table 1 Panel (b). The first “reweight” column rescales the coefficients in the large and small firm columns such that they sum to the total and the second “sum” column replaces the aggregate change with the sum of the large and small coefficients. The model column reports the difference in white employment rates and the racial employment gap between a low-productivity steady state and the baseline high-productivity steady state. Low productivity is 0.011 log points lower, chosen such that the difference in the white employment rate in the first row matches between data and model. The units are percentage points.

tween steady states. The model overestimates the negative impact on employment for Black workers, but captures that the decrease in employment is driven by large firms.

The overall worsening of the employment gap in the model happens because changes in hiring policy lead Black workers’ over-representation at large firms to decrease and under-representation at small firms to worsen. The change in hiring policy in the model can be understood through the cost minimization solution in equation 24, which acts as a free entry condition. A decrease in productivity leads to lower labor demand, which leads $c_v(z)/q(\theta)$ to fall. This has heterogeneous effects on hiring policy due to size differences and racial disparities in hiring.

Firm size matters directly due to differences in hiring strategies. Because large firms are able to post vacancies relatively cheaply, $c_v(L) < c_v(S)$, they choose a relatively high-selectivity strategy. When lower market tightness leads to a reduction in vacancy costs, this has a smaller effect on large-firm average selectivity, reflecting diminishing returns to recruiting though higher selectivity. Since white workers capture relatively more matches at

small firms, they are especially exposed to the change in hiring strategy.

The change in selectivity creates feedback effects between small and large firms. White workers are especially exposed to the higher selectivity at small firms, leading workers' outside options, Ω_g , to become more similar. This increases wage pressures for Black workers, especially at large firms where Black workers were employed more. Higher wage pressures lead large firms to adjust their hiring of Black workers more and this effect is amplified by the bias of matching rates towards incumbent employment.²³

5.2. Role of racial disparities

The main results show that the racial employment gap worsens when aggregate labor market conditions deteriorate, and that this is primarily driven by adjustments in hiring at large firms. In this section, I evaluate how each of the three channels of racial disparities embedded in the model—biased matching rates, taste-based discrimination, and statistical discrimination—contribute to the magnitude of the change in the employment gap and the degree to which it is driven by large firms. To do so, I repeat the same steady state comparison with four alternative calibrations of the model.

The first (“exogenous bias”) replaces the three channels of bias with a minimal framework in which Black workers receive fewer matches at large and small firms for purely exogenous reasons. The share of vacancies that match with Black workers at each type of firm, λ_{Bz} , become calibrated parameters. I set taste-based and statistical discrimination parameters to zero and I calibrate the exogenous vacancy shares such that the model matches the racial employment gap and the difference in large-firm shares. Appendix 7.3 reports parameters and moments. I compare a low-productivity steady state such that the change in the white employment rate is the same as the baseline. In the exogenous bias model, the

²³These spillovers are amplified when firms pay competitive wages to individuals, which leads to intra-firm wage dispersion. Online Appendix B.2 reports additional results when the amount of within-firm wage dispersion is restricted.

change in the employment gap is smaller but it is entirely driven by large firms, consistent with the patterns from the data.

Table 8: Steady state comparison, alternative racial disparity calibrations

	Baseline	Exogenous bias	No matching distortion	No statistical disc.	No taste disc.
Δ White employment rate	-1.49	-1.49	-1.49	-1.49	-1.49
Large	-1.04	-1.06	-1.09	-1.02	-1.04
Small	-0.45	-0.42	-0.40	-0.47	-0.45
Δ Employment gap	-1.21	-0.31	-1.27	-2.24	-1.03
Large	-1.21	-0.40	-0.81	-2.14	-1.00
Small	0.01	0.09	-0.46	-0.10	-0.03

This table shows the change in the white employment rate and racial employment gap between steady states. The first column uses the baseline calibration. The second column uses the calibration with exogenously lower matching rates for Black workers. The remaining columns use the calibrations with each channel of racial disparities turned off. The units are percentage points. The low productivity value is chosen such that the change in white employment in the first row is the same across calibrations.

Next, I evaluate the importance of the three channels of racial disparities in the baseline model by recalibrating parameters with each channel turned off.²⁴ Using these alternative calibrations, I repeat the steady state comparison. Without the bias in matching rates towards incumbent employment (“no matching distortion”), the change in employment gap is similar to baseline but the composition is much more shifted towards small firms than large. Without statistical discrimination, the change in the employment gap is most severe. It is still driven by large firms but the gap worsens by an additional percentage point relative to the baseline case. Meanwhile, the change in employment gap without taste-based discrimination is most similar to baseline, though less severe at large firms.

Overall, these exercises reveal that the form of racial disparities in hiring is influential for how employment changes when economic conditions worsen. Black workers are especially worse off when taste-based discrimination is the prevailing barrier, rather than statistical. Exogenous differences in matching rates yield the smallest change in the racial employment

²⁴Full details on the targeted moments, model fit, and parameters are provided in Appendix 7.3.

gap. However, bias in matching rates, whether exogenously imposed or modeled endogenously with incumbent employment, is important for the outsized role that large firms play in the employment gap change.

6. Conclusion

This paper examines the interactions between firm types and the Black-white employment gap when aggregate labor market conditions worsen. Using micro data, within individuals, Black employment at large firms is more responsive to changes in the state unemployment rate. This pattern is apparent in hiring as well.

I show that a flexible model of racial disparities in the hiring process can generate both the sorting of Black workers towards large firms and the disproportionate responsiveness of Black employment to aggregate productivity changes. Rather than assuming one form of racial disparities, I allow for three channels and use data to tease out the relative magnitudes of each. In the calibrated model, removing statistical discrimination reverses the employment gap but maintains the strong sorting of Black workers towards large firms. Removing incumbency bias in matching rates reduces sorting the most.

A permanent decrease in productivity leads to a steady state with a wider racial employment gap because firms become more selective about hiring, which disproportionately affects Black workers. Workers' outside options adjust, amplifying the effects for large firms especially. The relative adjustments of large and small firms depend on the source of racial hiring disparities. Distortions in matching rates towards incumbent employment are especially important for the degree to which large firms affect the total change in employment.

The spillover effects between small and large firms suggest that regulating only the biggest players will miss some of these economy-wide effects. They also suggest that although a tighter labor market can improve employment outcomes for marginalized groups more, the

full gains of the tight economy are not felt equally, with potential differences in the pass-through of wages.

7. Appendix

7.1. Empirical Appendix

SIPP Data definitions My sample includes about 215,000 individuals, whom I observe for an average of 28 months. Table 9 reports summary statistics.

I define occupations at the two-digit level. The occupation classification system changes between the 2001 and 2004 panels. Rather than impose an imperfect mapping, I define a set of 17 categories for the first half of the sample and a set of 22 categories for the second. For the 1996 and 2001 panels the categories are: executive, administrative, and managerial; management-related; professional specialty; technicians and related; sales; administrative support; housekeeping and cleaning; protective service; farm operations and managers; other agriculture; mechanics and repairers; construction trades; extractive occupations; precision production; machine operators; transportation. For the 2004 and 2008 panels the categories are: management; business and financial; computer and math; engineering; science; social work; legal; education; artists, entertainment, and media; health; health support; law enforcement; food service; maintenance; services; retail; administrative and postal; agriculture; extraction and trades; maintenance and installation; production; transportation.

I define industries at the two-digit level: (11) agriculture, (21) forestry, fishing and hunting; mining, quarrying, and oil and gas extraction; (22) utilities; (23) construction; (31-33) manufacturing; (42) wholesale trade; (44-45) retail trade; (48-49) transportation and warehousing; (51) information; (52-53) finance, insurance, and real estate; (55-56) management, administrative support, and waste management services; (61) educational services; (62) health care and social assistance; (71) arts, entertainment, and recreation; (72) accom-

Table 9: Summary statistics

	Black		White	
	Mean	SD	Mean	SD
Demographics				
Age (years)	39.8	12.9	42.8	13.7
Women	55.1	49.7	48.5	50.0
Married	38.6	48.7	62.3	48.5
College	17.7	38.2	31.7	46.5
Metro area	86.6	34.1	76.3	42.5
State unemployment	6.0	2.1	5.8	2.1
Employed				
Any	75.9	42.8	83.9	36.8
Large	39.5	48.9	37.3	48.4
Small	14.9	35.6	21.9	41.3
Government	16.3	37.0	13.2	33.9
Self-employment	4.6	20.8	10.5	30.6
Hires				
Any	3.1	17.2	2.7	16.1
Large	1.9	13.5	1.3	11.4
Small	0.9	9.4	1.0	10.1
Separations				
Any	2.9	16.7	2.6	15.9
Large	1.7	12.8	1.2	11.1
Small	0.8	8.9	1.0	9.8

The table reports means and standard deviations from the SIPP. The units are percentage points unless otherwise specified.

modation and food services; (81) other services; (92) public administration.

I define education in four bins: less than high school degree, high school degree or equivalent, some college, and college degree.

I construct a measure of firm size using three survey questions: “About how many persons are employed by ...’s employer at the location where ... works?” (tempsiz), “Does ...’s employer operate in more than one location?” (eemploc), and “About how many persons were employed by ...’s employer at ALL LOCATIONS together” (tempall). I choose 100

employees as the cutoff for large firms because it is available across waves even though the bins change over time. For all panels before 2008, there were three bins for both establishment and firm size with the largest being 100 or more. These bins were used in the 2008 panel as well until the 11th wave, when the bins were expanded to include eight bins for establishment size (three with 100 or fewer) and six bins for firm size (two with 100 or fewer). The number of individuals reporting working for the same employer but at a different size increases substantially in the 11th wave and remains elevated for the rest of the sample. For this reason, I exclude the final six waves of the panel.

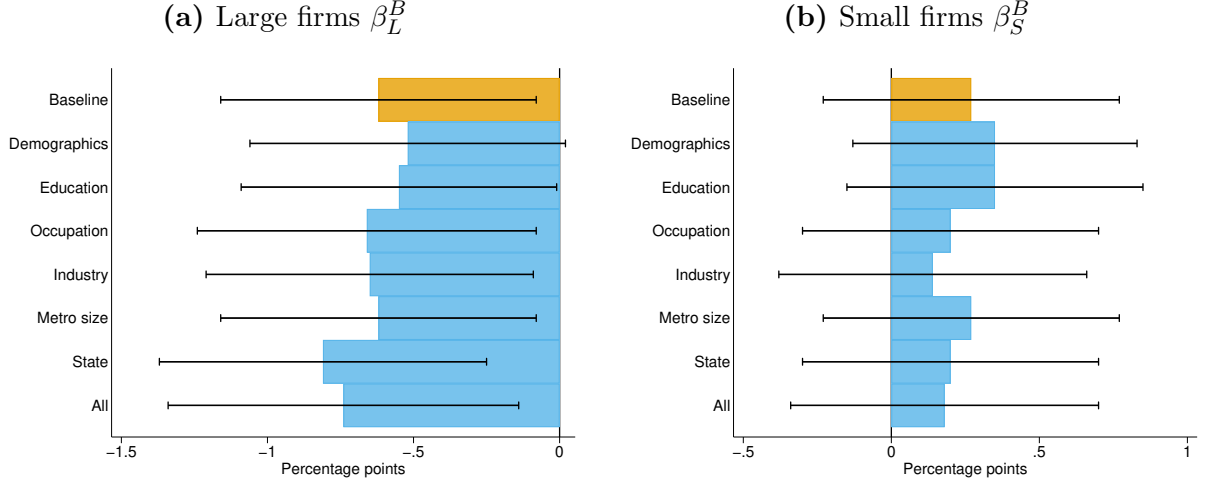
Robustness exercises In this section, I modify the primary regression specification to include interactions between worker characteristics and the state unemployment rate in addition to the Black dummy variable interaction,

$$E_{ijt} = \alpha_{ij} + \beta_j \text{UR}_t + \beta_j^B \text{Black} \times \text{UR}_t + \Gamma X_{it} \times \text{UR}_t + \epsilon_{ijt}. \quad (25)$$

This allows for the possibility that the change in the employment rate for Black workers is more responsive due to differences in industry, education, or other characteristics. Figure 2 reports the coefficient estimates with different control variables interacted. The estimates are relatively stable across the different specifications.

Alternative sample In the data, the relative responsiveness of Black employment to aggregate conditions may happen through firms expanding or contracting employment and/or firm entry and exit. The model focuses on just the intensive margin. I report below the main regression results for employment, dropping individuals who report losing their job due to firm closure or sale, to exclude the firm exit margin. This leads to dropping 1.4 percent of worker-month observations or 1.1 percent of workers. These shares are similar by race—I drop 1.3 percent of worker-month observations or 1.0 percent of individuals for

Figure 2: Regression estimates with additional control interaction terms



The figure reports the coefficient estimates for the interaction between the Black dummy variable and the state unemployment rate from equation 25 with varying controls X_{ijt} . Demographics includes age and age-squared, both standardized, and marital status interacted with gender.

Black workers.

7.2. Model Appendix: Wage details

The firm's value at the time of bargaining is given by

$$\begin{aligned}
 D_t(\{\tilde{n}_g\}, \{h_g\}, \{\bar{x}_g\}, z) &= a_t z (n')^\alpha - \sum_g \left(\tilde{n}_g (w_t^n(n', z, g) + \chi_{gz}) + h_g (w_t^h(x_g, n', z, g) + \chi_{gz}) \right) \\
 &\quad + \beta \mathbb{E}_t J_{t+1}(n'_B, n'_W, z) \\
 \text{s.t.} \\
 n' &= \sum_g n'_g \\
 n'_g &= \tilde{n}_g + \bar{x}_g h_g
 \end{aligned}$$

where $\tilde{n}_g = (1 - \delta)n_g$ is the number of non-separated workers from group g from the previous period and $h_g = \lambda_{gz} v q(\theta_t)(1 - F(x_g | p_{gz}))$ is the number of hires from group g with average expected productivity $\bar{x}_g$. The last line shows the mapping back to the law of motion in

Table 10: Employment fluctuations by firm type, excluding firm closure

(a) Individual controls					
	(1)	(2)	(3)	(4)	(5)
	All	Large	Small	Government	Self
Black	-4.08	2.64	-7.28	2.73	-2.07
	(0.23)	(0.31)	(0.25)	(0.22)	(0.17)
UR	-1.17	-0.51	-0.48	-0.06	-0.14
	(0.13)	(0.13)	(0.11)	(0.08)	(0.09)
Black $\times$ UR	-0.63	-0.43	0.26	-0.72	0.23
	(0.22)	(0.28)	(0.21)	(0.19)	(0.15)
R2	0.069	0.170	0.064	0.387	0.131
N	5,882,224	5,882,224	5,882,224	5,882,224	5,882,224
Black mean	75.91	39.43	14.80	16.43	4.59
White mean	83.93	37.24	21.73	13.36	10.58

(b) Individual fixed effects					
	(1)	(2)	(3)	(4)	(5)
	All	Large	Small	Government	Self
UR	-1.47	-1.28	-0.29	0.02	0.02
	(0.26)	(0.16)	(0.11)	(0.07)	(0.06)
Black $\times$ UR	-0.61	-0.61	0.29	-0.33	0.15
	(0.28)	(0.27)	(0.25)	(0.16)	(0.12)
R2	0.576	0.742	0.685	0.855	0.831
N	6,011,957	6,011,957	6,011,957	6,011,957	6,011,957
Black mean	75.91	39.43	14.80	16.43	4.59
White mean	83.93	37.24	21.73	13.36	10.58

The table reports differences in employment probability by race and macroeconomic conditions. The units are percentage points. UR is the state-level unemployment rate expressed in standard deviations from the mean. Panel (a) includes controls for age; age-squared; marital status interacted with gender; education; state; metro area size; modal industry and occupation. Panel (b) includes individual fixed effects. Standard errors are clustered by person and month.

equation (6).

I take the marginal surplus for each group, $D_{t,\tilde{n}_g}$ and D_{t,h_g} , where the arguments of $D()$

are omitted to ease notation.

$$\begin{aligned}
D_{t,\tilde{n}_g} &= \alpha a_t z(n')^{\alpha-1} - w_t^n(n', z, g) - \chi_{gz} - \sum_k \left(\tilde{n}_k w_{t,n'}^n(n', z, k) + h_k w_{t,n'}^h(\bar{x}_k, n', z, k) \right) \\
&\quad + \beta(1 - \delta) \mathbb{E}_t D_{t+1, \tilde{n}_g} \\
D_{t,\tilde{h}_g} &= \bar{x}_g \alpha a_t z(n')^{\alpha-1} - w_t^h(\bar{x}_g, n', z, g) - \chi_{gz} - \bar{x}_g \sum_k \left(\tilde{n}_k w_{t,n'}^n(n', z, k) + h_k w_{t,n'}^h(\bar{x}_k, n', z, k) \right) \\
&\quad + \beta(1 - \delta) \bar{x}_g \mathbb{E}_t D_{t+1, \tilde{n}_g}
\end{aligned}$$

The marginal surplus from the worker's side is given by

$$\begin{aligned}
V_t^n(n', g, z) - V_t^u(g) &= w_t^n(n', z, g) - (b + \Omega_t(g)) + \beta(1 - \delta) \mathbb{E}_t [V_{t+1}^n(n'', g, z) - V_{t+1}^u(g)] \\
V_t^h(\bar{x}_g, n', g, z) - V_t^u(g) &= w_t^h(\bar{x}_g, n', z, g) - (b + \Omega_t(g)) + \beta(1 - \delta) \bar{x}_g \mathbb{E}_t [V_{t+1}^n(n'', g, z) - V_{t+1}^u(g)]
\end{aligned}$$

Using the bargaining rules defined in equations (14) and (15),

$$\begin{aligned}
w_t^n(n', z, g) &= \phi \alpha a_t z(n')^{\alpha-1} - \phi \sum_k \left(\tilde{n}_k w_{t,n'}^n(n', z, k) + h_k w_{t,n'}^h(\bar{x}_k, n', z, k) \right) \\
&\quad - \phi \chi_{gz} + (1 - \phi)(b + \Omega_t(g)) \\
w_t^h(\bar{x}_g, n', z, g) &= \bar{x}_g \phi \alpha a_t z(n')^{\alpha-1} - \bar{x}_g \phi \sum_k \left(\tilde{n}_k w_{t,n'}^n(n', z, k) + h_k w_{t,n'}^h(\bar{x}_k, n', z, k) \right) \\
&\quad - \phi \chi_{gz} + (1 - \phi)(b + \Omega_t(g))
\end{aligned}$$

Notice that the relationship between new hire wages and existing worker wages is given by

$$w_t^h(\bar{x}_g, n', z, g) = \bar{x}_g w_t^n(n', z, g) + (1 - \bar{x}_g) [(1 - \phi)(b + \Omega_t(g)) - \phi \chi_{gz}]$$

which implies

$$w_{t,n'}^h(x_g, n', z, g) = \bar{x}_g w_{t,n'}^n(n', z, g)$$

Next, the wage gap between existing workers from the two groups is given by

$$w_t^n(n', z, B) - w_t^n(n', z, W) = (1 - \phi)(\Omega_t(B) - \Omega_t(W)) + \phi(\chi_{Wz} - \chi_{Bz})$$

which doesn't depend on the employment at the firm, and so $w_{n'}(n', z, B) = w_{n'}(n', z, W)$.

Using these observations, I can simplify the differential equation for $w_t^n(n', z, g)$,

$$w_t^n(n', z, g) = \phi a_t z (n')^{\alpha-1} - \phi n' w_{t,n'}^n(n', z, g) + (1 - \phi)(b + \Omega_t(g)) - \phi \chi_{gz}$$

Solving this differential equation gives the following equilibrium wages

$$\begin{aligned} w_t^n(n', z, g) &= \frac{\alpha\phi}{1 - \phi + \alpha\phi} a_t z (n')^{\alpha-1} + (1 - \phi)(b + \Omega_t(g)) - \phi \chi_{gz} \\ w_t^h(\bar{x}_g, n', z, g) &= \bar{x}_g \frac{\alpha\phi}{1 - \phi + \alpha\phi} a_t z (n')^{\alpha-1} + (1 - \phi)(b + \Omega_t(g)) - \phi \chi_{gz} \end{aligned}$$

7.3. Alternative calibrations of racial disparities

In the main calibration, I include three forms of potential racial disparities in hiring and discipline the channels using different moments from the data. To evaluate how important the assumed form of discrimination is for the counterfactual results, I recalibrate the model under four alternative scenarios. First, I assume that Black workers receive fewer matches than white workers for purely exogenous reasons. In this specification, λ_{BL} and λ_{BS} are exogenous parameters that I calibrate to match the gap in employment rates and sorting towards large firms directly, rather than microfounding with discrimination and incumbency bias. Next, I recalibrate the full specification with one form of discrimination excluded.

When omitting bias in vacancy-matching, I remove the target for the contact gap at large firms, since this will now be equal by assumption. When omitting either statistical or taste-based discrimination, I target only the gaps in employment and sorting rates, and leave the gaps in separation rates as untargeted.

Table 11 shows the fitted parameters across the alternative calibrations. The parameters that affect racial gaps are the ones that change the most across versions, while the aggregate parameters are relatively stable. Table 12 shows the model fit for the moments related to racial disparities. All specifications target the overall employment gap and the gap in sorting towards large firms and all calibrations hit these targets exactly. The version with exogenous bias in matching rates does not target any other moments. It captures the overall negative job-finding gap and positive separation gap, but it attributes too much of these gaps to large firms. The contact gap, or ratio of matches for white workers relative to Black workers is higher than the estimates from the correspondence literature, but the overall average is closer than in the main specification. The model without incumbency bias in matching rates looks most similar to the baseline calibration, except here the gap in job-finding rates is entirely coming from hiring decisions rather than match distortions. Without statistical discrimination, separation gaps become negative because firms are more selective about hiring Black workers and without noise distorting these decisions, this makes Black workers more positively selected. Finally, without taste discrimination, job-finding gaps are less severe but separation gaps are more severe, reflecting the stronger statistical discrimination channel.

Table 11: Fitted parameters across alternative assumptions about racial disparities

Parameter	Meaning	Baseline	Exogenous bias	No matching bias	No statistical disc.	No taste disc.
ζ	Matching scale	0.640	0.645	0.640	0.638	0.641
μ	Number firms/worker	0.009	0.009	0.009	0.009	0.009
b	Flow value unemp	0.980	0.981	0.979	0.976	0.981
ϕ	Bargaining power	0.353	0.352	0.357	0.362	0.349
δ	Exog. separation	0.024	0.024	0.024	0.024	0.024
$c_v(L)$	Vacancy cost	0.013	0.013	0.013	0.013	0.013
$c_v(S)$	Vacancy cost	0.181	0.187	0.171	0.184	0.171
$\frac{z_L}{z_S}$	Relative productivity	4.080	4.081	4.079	4.070	4.092
	Matching distortion					
λ	Incumbency bias	0.635	N/A	0.000	0.654	0.632
λ_{BL}	Large share	N/A	0.118	N/A	N/A	N/A
λ_{BS}	Small share	N/A	0.109	N/A	N/A	N/A
Δp_L	Signal gap, large	0.191	0.000	0.210	0.000	0.214
Δp_S	Signal gap, small	0.284	0.000	0.413	0.000	0.406
$\Delta \chi_L$	Taste penalty, large	0.002	0.000	0.007	0.012	0.000
$\Delta \chi_S$	Taste penalty, small	0.010	0.000	0.027	0.035	0.000

Table 12: Moments across alternative calibrations

Moment	Data	Baseline	Exogenous bias	No matching bias	No statistical disc.	No taste disc.
Employment, ppt (B-W)						
Empl rate	-4.08	-4.08	-4.08	-4.08	-4.08	-4.08
Large-firm share	6.20	6.20	6.20	6.20	6.20	6.20
Job-finding, ppt (B-W)						
Large	0.10	0.00	-0.59	0.00	-0.87	0.11
Small	-2.23	-1.23	-1.66	-1.23	-2.07	-0.80
Separation, ppt (B-W)						
Large	0.40	0.40	0.11	0.40	-0.02	0.45
Small	0.51	0.51	0.05	0.51	-0.39	0.98
Contact gap	0.36	0.13	0.43	0.00	0.13	0.13
Large firms	0.09	0.09	0.40	0.00	0.09	0.09

The figures in bold represent untargeted moments. The data moments are all calculated in the SIPP except for the external moments and contact gaps.

References

- Aaronson, Stephanie R., Wascher, William L., Daly, Mary C., & Wilcox, David W. 2019. Okun Revisited: Who Benefits Most from a Strong Economy? *Brookings Papers on Economic Activity*, **2019**(Spring), 333–404.
- Acemoglu, Daron, & Hawkins, William B. 2014. Search with Multi-Worker Firms. *Theoretical Economics*, **9**(3), 583–628.
- Axtell, Robert L. 2001. Zipf Distribution of U.S. Firm Sizes. *Science*, **293**(5536), 1818–1820.
- Banerjee, Rupa, Reitz, Jeffrey G., & Oreopoulos, Phil. 2018. Do Large Employers Treat Racial Minorities More Fairly? An Analysis of Canadian Field Experiment Data. *Canadian Public Policy*, **44**(1), 1–12.
- Barron, John M., Berger, Mark C., & Black, Dan A. 1997. Employer Search, Training, and Vacancy Duration. *Economic Inquiry*, **XXXV**, 167–192.
- Bartscher, Alina K., Kuhn, Moritz, Schularick, Moritz, & Wachtel, Paul. 2021. Monetary Policy and Racial Inequality. *Federal Reserve Bank of New York Staff Reports*.
- Baydur, Ismail. 2017. Worker Selection, Hiring, and Vacancies. *American Economic Journal: Macroeconomics*, **9**(1), 88–127.
- Bergman, Nittai, Matsa, David A., & Weber, Michael. 2020. *Heterogeneous Labor Market Effects of Monetary Policy*. Chicago Booth Research Paper No. 21-02.
- Bertrand, Marianne, & Mullainathan, Sendhil. 2004. Are Emily and Greg More Employable than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination. *The American Economic Review*, **94**(4), 991–1013.
- Black, Dan A. 1995. Discrimination in an Equilibrium Search Model. *Journal of Labor Economics*, **13**(2), 309–334.
- Boulware, Karl David, & Kuttner, Kenneth N. 2019. Labor Market Conditions and Discrimination: Is There a Link? *AEA Papers and Proceedings*, **109**, 166–170.

- Cajner, Tomaz, Radler, Tyler, Ratner, David, & Vidangos, Ivan. 2017. *Racial Gaps in Labor Market Outcomes in the Last Four Decades and over the Business Cycle*. Finance and Economics Discussion Series 2017-071.
- Carpenter, Seth B., & Rodgers III, William M. 2004. The Disparate Labor Market Impacts of Monetary Policy. *Journal of Policy Analysis and Management*, **23**(4), 813–830.
- Cavounidis, Costas, Lang, Kevin, & Weinstein, Russell. 2024. The Boss Is Watching: How Monitoring Decisions Hurt Black Workers. *The Economic Journal*, **134**(658), 485–514.
- Charles, Kerwin Kofi, & Guryan, Jonathan. 2008. Prejudice and Wages: An Empirical Assessment of Becker’s The Economics of Discrimination. *Journal of Political Economy*, **116**(5), 773–809.
- Couch, Kenneth A, & Fairlie, Robert. 2010. Last Hired, First Fired? Black-White Unemployment and the Business Cycle. *Demography*, **47**(1), 227–247.
- De, Kuhelika, Compton, Ryan A., Giedeman, Daniel C., & Hoover, Gary A. 2021. Macroeconomic Shocks and Racial Labor Market Differences. *Southern Economic Journal*, **88**(2), 680–704.
- Derenoncourt, Ellora, Kim, Chi Hyun, Kuhn, Moritz, & Schularick, Moritz. 2024. Wealth of Two Nations: The U.S. Racial Wealth Gap, 1860–2020. *The Quarterly Journal of Economics*, **139**(2), 693–750.
- Elsby, Michael W. L, & Michaels, Ryan. 2013. Marginal Jobs, Heterogeneous Firms, and Unemployment Flows. *American Economic Journal: Macroeconomics*, **5**(1), 1–48.
- Flood, Sarah, King, Miriam, Rodgers, Renae, Ruggles, Steven, & Warren, J. Robert. 2022. *Integrated Public Use Microdata Series, Current Population Survey: Version 10.0 [Dataset]*.
- Forsythe, Eliza, & Wu, Jhih-Chian. 2021. Explaining Demographic Heterogeneity in Cyclical Unemployment. *Labour Economics*, **69**, 101955.
- Fujita, Shigeru, & Nakajima, Makoto. 2016. Worker Flows and Job Flows: A Quantitative

- Investigation. *Review of Economic Dynamics*, **22**(Oct.), 1–20.
- Fujita, Shigeru, Nekarda, Christopher J., & Ramey, Garey. 2007. *The Cyclicalities of Worker Flows: New Evidence from the SIPP*. Tech. rept. FRB of Philadelphia Working Paper No.07-5.
- Haltiwanger, John C., Hyatt, Henry R., Kahn, Lisa B., & McEntarfer, Erika. 2018. Cyclical Job Ladders by Firm Size and Firm Wage. *American Economic Journal: Macroeconomics*, **10**(2), 52–85.
- Holzer, Harry J. 1998. Why Do Small Establishments Hire Fewer Blacks Than Large Ones? *The Journal of Human Resources*, **33**(4), 896–914.
- Hoynes, Hilary, Miller, Douglas L., & Schaller, Jessamyn. 2012 (June). Who Suffers during Recessions? *Pages 27–48 of: Journal of Economic Perspectives*, vol. 26.
- Jovanovic, Boyan. 1979. Job Matching and the Theory of Turnover. *Journal of Political Economy*, **87**(5), 972–990.
- Kline, Patrick, Rose, Evan K., & Walters, Christopher R. 2022. Systemic Discrimination Among Large U.S. Employers. *The Quarterly Journal of Economics*, **137**(4), 1963–2036.
- Kuhn, Florian, & Chanci, Luis. 2021. *Effects of Hiring Discrimination over the Business Cycle*. Tech. rept.
- Lang, Kevin, & Lehmann, Jee-Yeon K. 2012. Racial Discrimination in the Labor Market: Theory and Empirics. *Journal of Economic Literature*, **50**(4), 959–1006.
- Lee, Munseob, Macaluso, Claudia, & Schwartzman, Felipe. 2025. *Minority Unemployment, Inflation, and Monetary Policy*. Working Paper.
- Menzio, Guido, & Shi, Shouyong. 2011. Efficient Search on the Job and the Business Cycle. *Journal of Political Economy*, **119**(3), 468–510.
- Miller, Conrad. 2017. The Persistent Effect of Temporary Affirmative Action. *American Economic Journal: Applied Economics*, **9**(3), 152–190.
- Miller, Conrad, & Schmutte, Ian. 2023. The Dynamic Effects of Co-Racial Hiring.

- Moore, Jeffrey C. 2008. *Seam Bias in the 2004 SIPP Panel: Much Improved, but Much Bias Still Remains*. Tech. rept.
- Morgan, John, & Várdy, Felix. 2009. Diversity in the Workplace. *American Economic Review*, **99**(1), 472–485.
- Moscarini, Giuseppe, & Postel-Vinay, Fabien. 2012. The Contribution of Large and Small Employers to Job Creation in Times of High and Low Unemployment. *American Economic Review*, **102**(6), 2509–2539.
- Petrongolo, Barbara, & Pissarides, Christopher A. 2001. Looking into the Black Box: A Survey of the Matching Function. *Journal of Economic Literature*, **39**(2), 390–431.
- Quillian, Lincoln, Pager, Devah, Hexel, Ole, & Midtbøen, Arnfinn H. 2017. Meta-Analysis of Field Experiments Shows No Change in Racial Discrimination in Hiring over Time. *Proceedings of the National Academy of Sciences of the United States of America*, **114**(41), 10870–10875.
- Sorkin, Isaac. 2025 (June). *Quantifying Racial Disparities Using Consecutive Employment Spells*. NBER Working Paper 33946. National Bureau of Economic Research.
- Stole, Lars A., & Zwiebel, Jeffrey. 1996. Intra-Firm Bargaining under Non-binding Contracts. *The Review of Economic Studies*, **63**(3), 375–410.
- Thorbecke, Willem. 2001. Estimating the Effects of Disinflationary Monetary Policy on Minorities. *Journal of Policy Modeling*, **23**(1), 51–66.
- US Census Bureau. 1996. *Survey of Income and Program Participation*.
- US Census Bureau. 2023. *QWI Explorer*.
- Villena-Roldan, Benjamin. 2012. *Aggregate Implications of Employer Search and Recruiting Selection*. University of Chile Center for Applied Economics Working Paper 271.
- Zavodny, Madeline, & Zha, Tao. 2000. Monetary Policy and Racial Unemployment Rates. *Economic Review- Federal Reserve Bank of Atlanta*, **85**(4), 1–59.

Online Appendix

A. Additional Empirical Results

A.1. Heterogeneity by gender

This section reports the main empirical results by gender. Tables [A.1](#) and [A.2](#) show the results from Table [1](#) by gender. The results for men are weaker than the results for women, and the individual fixed effects are more important.

Table A.1: Employment fluctuations by firm type, men

(a) Individual controls					
	(1)	(2)	(3)	(4)	(5)
	All	Large	Small	Government	Self
Black	-5.29	1.45	-6.61	2.16	-2.36
	(0.32)	(0.46)	(0.37)	(0.31)	(0.28)
UR	-1.75	-0.85	-0.82	0.09	-0.19
	(0.14)	(0.17)	(0.16)	(0.10)	(0.13)
Black $\times$ UR	-0.43	-0.12	0.03	-0.67	0.35
	(0.32)	(0.42)	(0.32)	(0.27)	(0.25)
R2	0.082	0.185	0.068	0.408	0.142
N	2,964,624	2,964,624	2,964,624	2,964,624	2,964,624
Black mean	76.70	39.45	15.74	14.60	6.17
White mean	86.90	38.83	22.23	11.53	13.43

(b) Individual fixed effects					
	(1)	(2)	(3)	(4)	(5)
	All	Large	Small	Government	Self
UR	-1.63	-1.41	-0.34	0.12	-0.06
	(0.30)	(0.18)	(0.14)	(0.08)	(0.09)
Black $\times$ UR	-0.57	-0.65	0.21	-0.54	0.45
	(0.38)	(0.41)	(0.35)	(0.22)	(0.21)
R2	0.563	0.749	0.686	0.868	0.847
N	3,019,437	3,019,437	3,019,437	3,019,437	3,019,437
Black mean	76.70	39.45	15.74	14.60	6.17
White mean	86.90	38.83	22.23	11.53	13.43

The table reports differences in employment probability by race and macroeconomic conditions for men. The units are percentage points. UR is the state-level unemployment rate expressed in standard deviations from the mean. Panel (a) includes controls for age; age-squared; marital status interacted with gender; education; state; metro area size; modal industry and occupation. Panel (b) includes individual fixed effects. Standard errors are clustered by person and month.

Table A.2: Employment fluctuations by firm type, women

(a) Individual controls					
	(1) All	(2) Large	(3) Small	(4) Government	(5) Self
Black	-2.88 (0.31)	3.56 (0.41)	-7.46 (0.32)	3.11 (0.31)	-1.86 (0.19)
UR	-0.62 (0.19)	-0.20 (0.18)	-0.13 (0.15)	-0.24 (0.12)	-0.05 (0.11)
Black $\times$ UR	-0.97 (0.29)	-0.81 (0.37)	0.34 (0.28)	-0.73 (0.27)	0.15 (0.17)
R2	0.056	0.158	0.070	0.375	0.114
N	3,002,944	3,002,944	3,002,944	3,002,944	3,002,944
Black mean	75.20	39.48	14.18	17.72	3.24
White mean	80.67	35.67	21.48	14.98	7.37

(b) Individual fixed effects					
	(1) All	(2) Large	(3) Small	(4) Government	(5) Self
UR	-1.35 (0.30)	-1.18 (0.20)	-0.26 (0.14)	-0.07 (0.11)	0.10 (0.07)
Black $\times$ UR	-0.74 (0.38)	-0.62 (0.35)	0.30 (0.31)	-0.15 (0.23)	-0.09 (0.14)
R2	0.577	0.729	0.676	0.843	0.793
N	3,077,848	3,077,848	3,077,848	3,077,848	3,077,848
Black mean	75.20	39.48	14.18	17.72	3.24
White mean	80.67	35.67	21.48	14.98	7.37

The table reports differences in employment probability by race and macroeconomic conditions for women. The units are percentage points. UR is the state-level unemployment rate expressed in standard deviations from the mean. Panel (a) includes controls for age; age-squared; marital status interacted with gender; education; state; metro area size; modal industry and occupation. Panel (b) includes individual fixed effects. Standard errors are clustered by person and month.

Tables [A.3](#) and [A.4](#) report the results from Table [2](#) separately by gender. Again, the results are generally stronger for women. For men, the Black-unemployment interaction coefficients become positive for job-finding (though not for hiring). Again, the attenuation is smaller at large firms, indicating that large firms are still relatively more important for the gap in job-finding rates in higher unemployment periods.

Table A.3: Job inflows by firm type, men

(a) Hires (percent of population)						
	(1)	(2)	(3)	(4)	(5)	(6)
	All	Large	Small	All	Large	Small
Black	0.08	0.28	-0.26			
	(0.06)	(0.04)	(0.03)			
UR	-0.05	-0.06	-0.02	-0.08	-0.12	0.01
	(0.03)	(0.02)	(0.02)	(0.09)	(0.04)	(0.03)
Black $\times$ UR	-0.06	-0.04	-0.01	-0.22	-0.15	-0.10
	(0.05)	(0.04)	(0.03)	(0.10)	(0.09)	(0.05)
R2	0.012	0.007	0.007	0.079	0.072	0.072
N	2,964,624	2,964,624	2,964,624	3,019,437	3,019,437	3,019,437
Black mean	3.02	1.75	0.96	3.02	1.75	0.96
White mean	2.54	1.24	1.04	2.54	1.24	1.04
Individual FE				X	X	X

(b) Job-finding						
	(1)	(2)	(3)	(4)	(5)	(6)
	All	Large	Small	All	Large	Small
Black	-3.25	-0.62	-2.49	-7.82	-2.32	-5.15
	(0.31)	(0.21)	(0.17)	(0.63)	(0.46)	(0.36)
UR	-1.69	-0.91	-0.65	-5.42	-2.91	-2.17
	(0.21)	(0.11)	(0.10)	(0.28)	(0.19)	(0.17)
Black $\times$ UR	0.42	0.13	0.28	2.00	0.62	1.31
	(0.25)	(0.19)	(0.13)	(0.46)	(0.36)	(0.25)
R2	0.037	0.027	0.021	0.036	0.028	0.024
N	372,813	372,813	372,813	116,110	116,110	116,110
Black mean	9.88	5.46	3.05	16.09	9.18	5.10
White mean	13.41	6.16	5.26	24.19	11.59	10.02
Unemployed sample				X	X	X

The table reports differences in hiring rates by race and macroeconomic conditions for men. The units are percentage points. UR is the state-level unemployment rate expressed in standard deviations from the mean. Panel (a) includes all individuals and panel (b) restricts to non-employed or unemployed individuals. All columns without fixed effects include controls for age; age-squared; marital status interacted with gender; education; state; metro area size; modal industry and occupation. Standard errors are clustered by person and month.

Table A.4: Job inflows by firm type, women

(a) Hires (percent of population)						
	(1)	(2)	(3)	(4)	(5)	(6)
	All	Large	Small	All	Large	Small
Black	-0.02	0.34	-0.34			
	(0.05)	(0.04)	(0.03)			
UR	-0.17	-0.10	-0.07	-0.21	-0.18	-0.05
	(0.04)	(0.02)	(0.01)	(0.12)	(0.06)	(0.04)
Black $\times$ UR	-0.09	-0.13	0.03	-0.23	-0.14	-0.07
	(0.04)	(0.03)	(0.02)	(0.09)	(0.07)	(0.05)
R2	0.010	0.007	0.005	0.070	0.066	0.064
N	3,002,944	3,002,944	3,002,944	3,077,848	3,077,848	3,077,848
Black mean	3.07	1.93	0.85	3.07	1.93	0.85
White mean	2.77	1.39	1.03	2.77	1.39	1.03
Individual FE				X	X	X

(b) Job-finding						
	(1)	(2)	(3)	(4)	(5)	(6)
	All	Large	Small	All	Large	Small
Black	-1.28	0.57	-1.62	-8.27	-2.32	-5.02
	(0.22)	(0.16)	(0.12)	(0.53)	(0.45)	(0.32)
UR	-0.75	-0.45	-0.27	-5.16	-2.87	-1.81
	(0.19)	(0.10)	(0.07)	(0.37)	(0.24)	(0.18)
Black $\times$ UR	-0.60	-0.58	0.01	1.25	0.01	1.10
	(0.17)	(0.14)	(0.09)	(0.37)	(0.31)	(0.20)
R2	0.019	0.019	0.010	0.040	0.026	0.023
N	533,258	533,258	533,258	107,092	107,092	107,092
Black mean	9.52	5.75	2.59	17.25	10.93	4.60
White mean	10.29	4.85	3.64	25.07	12.86	8.67
Unemployed sample				X	X	X

The table reports differences in hiring rates by race and macroeconomic conditions for women. The units are percentage points. UR is the state-level unemployment rate expressed in standard deviations from the mean. Panel (a) includes all individuals and panel (b) restricts to non-employed or unemployed individuals. All columns without fixed effects include controls for age; age-squared; marital status interacted with gender; education; state; metro area size; modal industry and occupation. Standard errors are clustered by person and month.

Tables A.5 and A.6 report the results of Table 3 by gender. For men, we see a small worsening of the racial separation gap for separations with higher unemployment, consistent with previous literature that focused on men, such as Couch & Fairlie (2010). The point estimates are not statistically significant. We see the opposite pattern for women, which is driving the overall pattern seen in Table 3.

Table A.5: Separation rates by firm type, men

	(1)	(2)	(3)	(4)	(5)	(6)
	All	Large	Small	All	Large	Small
Black	0.38	0.46	0.63	0.38	0.49	0.58
	(0.07)	(0.09)	(0.17)	(0.05)	(0.06)	(0.12)
UR	0.05	0.00	0.17	0.21	0.21	0.41
	(0.04)	(0.04)	(0.07)	(0.03)	(0.03)	(0.05)
Black $\times$ UR	0.06	0.04	0.01	0.07	0.08	0.01
	(0.07)	(0.09)	(0.16)	(0.05)	(0.07)	(0.12)
R2	0.013	0.013	0.011	0.007	0.007	0.007
N	2,471,350	1,171,446	654,400	2,588,226	1,171,446	654,400
Black mean	6.78	6.69	11.06	3.46	3.11	5.67
White mean	2.93	2.80	4.06	1.30	1.25	2.03
Involuntary				X	X	X

The table reports differences in separation rates by race and macroeconomic conditions for men. The units are percentage points. Columns (1)-(3) include all separations and columns (4)-(6) restrict the outcome variable to involuntary separations. UR is the state-level unemployment rate expressed in standard deviations from the mean. All columns include controls for age; age-squared; marital status interacted with gender; education; state; metro area size; modal industry and occupation. Standard errors are clustered by person and month.

A.2. Macro-level Empirical Patterns

Data The Quarterly Workforce Indicators (QWI) data are public-use aggregates made available by the U.S. Census Bureau from linked employer-employee microdata (US Census Bureau, 2023). They include quarterly employment, job creation, job destruction, average earnings, and other measures of employment flows that are available at several levels of disaggregation on both firm and worker characteristics. I focus on the statistics by race and

Table A.6: Separation rates by firm type, women

	(1)	(2)	(3)	(4)	(5)	(6)
	All	Large	Small	All	Large	Small
Black	0.14 (0.06)	0.32 (0.08)	0.30 (0.14)	0.32 (0.04)	0.40 (0.05)	0.44 (0.09)
UR	-0.11 (0.05)	-0.12 (0.05)	-0.18 (0.06)	0.14 (0.03)	0.14 (0.03)	0.17 (0.03)
Black $\times$ UR	-0.06 (0.05)	-0.18 (0.07)	0.10 (0.13)	-0.07 (0.03)	-0.06 (0.05)	-0.07 (0.08)
R2	0.012	0.012	0.011	0.004	0.005	0.005
N	2,355,302	1,119,732	629,286	2,462,015	1,119,732	629,286
Black mean	6.47	6.64	10.14	2.74	2.62	3.93
White mean	3.44	3.33	4.11	1.24	1.22	1.53
Involuntary				X	X	X

The table reports differences in separation rates by race and macroeconomic conditions for women. The units are percentage points. Columns (1)-(3) include all separations and columns (4)-(6) restrict the outcome variable to involuntary separations. UR is the state-level unemployment rate expressed in standard deviations from the mean. All columns include controls for age; age-squared; marital status interacted with gender; education; state; metro area size; modal industry and occupation. Standard errors are clustered by person and month.

firm size from 2001Q1 to 2019Q4.²⁵

Decomposition of employment volatility I start with the seasonally-adjusted time series of log employment for Black and white workers, and detrend it with an HP-filter. Consistent with the employment gap shown in Figure 1a, Black employment is more volatile than white employment, with a standard deviation of 0.016 log point, relative to 0.009 log point.²⁶ I use the time series of employment by firm size to decompose total employment deviations from trend into the contributions of large and small employers. First, I log-

²⁵The national QWI data start in 1993Q2, but national statistics are derived from state-level data, where the state coverage is increasing over time. In 1993, state-level data are available for roughly 20% of states (likely less by population), rising to over 80% by 2001. QWI statistics by race/ethnicity and firm size cannot be combined with any other worker characteristics, such as gender, age, or education, or with firm age.

²⁶Using a linear trend instead these standard deviations are 0.033 and 0.018 log points, respectively.

linearize total employment to decompose it into firm-size components,

$$\log e_{gt} - \log \bar{e}_{gt} = \sum_j (\log e_{jgt} - \log \bar{e}_{jgt}) \times \frac{\bar{e}_{jgt}}{\bar{e}_{gt}}, \quad (\text{A.1})$$

where $\bar{e}$ is the time trend in employment, g indexes race, t is quarter, and j is firm size. Figure A.1 show the contributions of small and large firms to detrended aggregate employment, which is the sum of both panels. Panel (a) shows that the total contribution of small firms is relatively similar for Black and white workers, while Panel (b) shows that large firms contribute significantly more to the fluctuations in employment for Black workers.

Figure A.1: Employment volatility by firm size



This figure shows the decomposition terms from Equation A.1 for HP-filtered log employment for Black and white workers. The two panels sum to the aggregate detrended employment across all sizes. Large firms have 250 or more employees. The units are log deviations from trend.

The difference in the contributions of large and small employers to Black and white employment fluctuations depends both on the size-specific employment fluctuations and the share of workers employed at each type of firm. Table A.7 Panel (a) reports the large-firm share of employment for Black and white workers. Large firms could contribute more to Black employment fluctuations because they are more volatile and Black workers are more likely to be employed by large firms, so they are more exposed to this volatility. To disentangle

this effect, I perform the following decomposition of the Black-white employment gap,

$$\tilde{e}_{Bt} - \tilde{e}_{Wt} = \underbrace{\sum_j \tilde{e}_{jWt} \times \left(\frac{\bar{e}_{jBt}}{\bar{e}_{Bt}} - \frac{\bar{e}_{jWt}}{\bar{e}_{Wt}} \right)}_{\text{Composition}} + \underbrace{\sum_j \left(\tilde{e}_{jBt} - \tilde{e}_{jWt} \right) \times \frac{\bar{e}_{jBt}}{\bar{e}_{Bt}}}_{\text{Excess volatility}}, \quad (\text{A.2})$$

where $\tilde{x}_t = \log(x_t) - \log(\bar{x}_t)$. I use this equation to perform a variance decomposition of the detrended employment gap between Black and white workers, which has a standard deviation of 0.009 log point. Results are reported in Table A.7 Panels (b) and (c). With the HP-filter trend, 3% of the variance can be explained by the composition effects, whereas the excess volatility of Black employment at large and small firms contribute 74% and 24%, respectively. To remove the scaling effects, that large firms have a higher Black share of employment and therefore are likely to make a bigger contribution, I first calculate the total excess volatility term in brackets in Equation A.2. Then I subtract the contribution that would be expected if the total excess volatility were distributed according to the employment shares from each term. The large firm excess volatility contribution is thus 6.4 percentage points higher than its proportional share of the total excess volatility term. Using a linear trend, composition makes a small negative contribution and the excess contribution of large firms is smaller but still sizeable at 5.3 percentage points.

I also apply Equation A.2 to total gaps in hires and separations, which have standard deviations of 0.025 and 0.023 log deviations from trend.²⁷ Composition effects make larger positive contributions for hires and separations than for total employment, though the error of these decompositions is higher for the HP-filter specification. The excess volatility of large firms is also larger for hires and separations across both specifications.

²⁷With the linear trend, these are both 0.05 log points.

Table A.7: Employment shares and variance decomposition

(a) Large firm share			(b) Variance decomposition, HP-filter			
	Black	White		Composition	Excess large	Error
Employment	69.4	53.9	Employment	3.3	6.4	-0.6
Hires	63.9	47.4	Hires	10.6	8.2	1.1
Separations	64.9	48.7	Separations	8.9	8.5	2.7

(c) Variance decomposition, linear trend			
	Composition	Excess large	Error
Employment	-1.1	5.3	-0.3
Hires	5.0	7.8	0.4
Separations	5.7	7.3	0.7

Panel (a) reports the large-firm (250 employees or more) share of employment, hires, and separations for Black and white workers in the QWI. Panels (b) and (c) report the variance decomposition using equation A.2 with different trend estimates. The excess volatility term is the contribution of large firms in excess of their relative share of employment, hires, or separations. The units are percentage points.

B. Removing within-firm wage dispersion

B.1. Model framework with wage equality

In the baseline model, workers of different races may be paid different wages, conditional on the same productivity, due to differences in outside options and pass-through of taste-based discrimination. In this section, I consider an extension where firms pay wages based on the average outside option of each group. Consider the wage for an incumbent worker,

$$\begin{aligned} \tilde{w}_t^n(n', z, g) = & \frac{\alpha\phi}{1 - \phi + \alpha\phi} a_t z (n')^{\alpha-1} \\ & + \sum_g ((1 - \phi)(b + \Omega_t(g)) - \phi\chi_{gz}) \frac{n_{gz}}{n_{Bz} + n_{Wz}}, \end{aligned} \quad (\text{A.3})$$

where the first term is invariant by race and the second term takes a weighted average of workers' outside options. For new hires, wages may differ by race both due to differences in expected productivity and due to differences in outside options. I allow workers to be paid

different wages on the basis of productivity, but not outside options,

$$\begin{aligned}\tilde{w}_t^h(\bar{x}_g, n', z, g) = & \bar{x}_g \frac{\alpha\phi}{1-\phi+\alpha\phi} a_t z (n')^{\alpha-1} \\ & + \sum_g ((1-\phi)(b+\Omega_t(g)) - \phi\chi_{gz}) \frac{h_{gz}}{h_{Bz} + h_{Wz}}.\end{aligned}\quad (\text{A.4})$$

To write this formulation more generally, let ι_z be the share of firms with productivity z that pay workers different wages by race. The worker value functions then become,

$$\begin{aligned}V_t^n(n, g, z) = & \iota_z w_t^n(n, z, g) + (1 - \iota_z) \tilde{w}_t^n(n, z, g) \\ & + \beta \mathbb{E}_t [V_{t+1}^u(g) + (1 - \delta)(V_{t+1}^n(n', g, z) - V_{t+1}^u(g))],\end{aligned}\quad (\text{A.5})$$

$$\begin{aligned}V_t^h(x, n, g, z) = & \iota_z w_t^h(x, n, z, g) + (1 - \iota_z) \tilde{w}_t^h(x, n, z, g) \\ & + \beta \mathbb{E}_t [V_{t+1}^u(g) + x(1 - \delta)(V_{t+1}^n(n', g, z) - V_{t+1}^u(g))],\end{aligned}\quad (\text{A.6})$$

where $\iota_z = 1$ is the baseline case and $\iota_z = 0$ is the other extreme where no wage discrimination is allowed.

While the worker value functions, and therefore worker outside options, $\Omega_t(g)$, adjust when wage discrimination is limited or removed, the firm's problem does not change. Hiring a worker of one group has the same marginal contribution to the firm's total wage bill that it did when wage discrimination was fully allowed. Firms do not internalize the general equilibrium effects their policies have on outside options.

Consider the partial equilibrium effects of this change by temporarily leaving firms' hiring decisions fixed. Increasing wages for Black workers and decreasing wages for white workers will lead to an increase in $\Omega_t(B)$ and a decrease in $\Omega_t(W)$. This is going to make it relatively more costly for firms to hire Black workers, leading them to substitute away from Black workers and towards white workers when we allow hiring decisions to adjust. Thus, wage equality leads to bigger adjustments on the employment margin.

B.2. Quantitative model with wage equality

I evaluate the importance of allowing wage dispersion by recalibrating the model under three alternative scenarios. First, I impose that firms must compensate workers according to the average outside option of all workers, such that the only wage differences that arise are due to productivity differences among new hires and incumbents. This is the framework outlined above. Second, I only allow wage dispersion at small firms, imposing that large firms must pay workers equal wages conditional on productivity. This allows for the possibility that large firms set wages differently due to stronger scrutiny around equal pay. Finally, I remove the possibility for wage dispersion in the simplified version of the model presented above in section 7.3. In this version of the model, Black workers face exogenously lower arrival rates of matches but there is no other discrimination and wages are restricted to be the same within firms, conditional on productivity.

Table A.8: Fitted parameters across calibrations without wage dispersion

Parameter	Meaning	Baseline	No wage dispersion	Only wage disp. at small firms	No wage disp, exog. bias
ζ	Matching scale	0.640	0.639	0.640	0.642
μ	Number firms/worker	0.009	0.008	0.008	0.009
b	Flow value unemp	0.980	0.952	0.967	0.981
ϕ	Bargaining power	0.353	0.520	0.425	0.352
δ	Exog. separation	0.024	0.023	0.023	0.024
$c_v(L)$	Vacancy cost	0.013	0.008	0.010	0.013
$c_v(S)$	Vacancy cost	0.181	0.342	0.262	0.186
$\frac{z_L}{z_S}$	Relative productivity	4.080	4.045	4.080	4.081
λ	Matching distortion	0.635	0.618	0.624	N/A
λ_{BL}	Matching distortion, large	N/A	N/A	N/A	0.128
λ_{BS}	Matching distortion, small	N/A	N/A	N/A	0.109
Δp_L	Signal gap, large	0.191	0.201	0.197	0.00
Δp_S	Signal gap, small	0.284	0.357	0.318	0.00
$\Delta \chi_L$	Taste penalty, large	0.002	0.001	0.001	0.00
$\Delta \chi_S$	Taste penalty, small	0.010	0.039	0.020	0.00

Table A.8 shows the fitted parameters across these three alternative calibrations. When

wages are restricted, the parameters governing discrimination generally become more severe. Signal noise goes up for Black workers, especially at small firms and the taste penalty rises at small firms. The model fit is similar, as shown in Table A.9.

Table A.9: Moments across calibrations without wage dispersion

(a) Aggregate moments					
Moment	Data	Baseline	No wage dispersion	Only wage disp. at small firms	No wage disp, exog. bias
Employment share					
Large	0.64	0.64	0.64	0.64	0.64
Job-finding rate, ppt					
Large	6.45	6.45	6.36	6.40	6.45
Small	4.78	4.78	4.86	4.82	4.78
Separation rate, ppt	3.17	3.17	3.17	3.17	3.17
Large	3.05	2.84	2.68	2.75	2.83
Small	4.07	3.78	4.18	3.98	3.78
External					
Hired share matches	0.20	0.20	0.20	0.20	0.20
Market tightness	0.72	0.72	0.72	0.72	0.72
Outside option	1.00	1.00	1.00	1.00	1.00
Ratio b to avg prod	0.73	0.73	0.73	0.73	0.73
(b) Racial gap moments					
Moment	Data	Baseline	No wage dispersion	Only wage disp. at small firms	No wage disp, exog. bias
Employment, ppt (B-W)					
Empl rate	-4.08	-4.08	-4.09	-4.09	-4.08
Large-firm share	6.20	6.20	6.20	6.20	6.20
Job-finding, ppt (B-W)					
Large	0.10	0.00	0.02	0.01	-0.59
Small	-2.23	-1.23	-1.37	-1.30	-1.66
Separation, ppt (B-W)					
Large	0.40	0.40	0.40	0.40	0.11
Small	0.51	0.51	0.51	0.51	0.05
Contact gap	0.36	0.13	0.12	0.12	0.43
Large firms	0.09	0.09	0.09	0.09	0.40

The figures in bold represent untargeted moments. The data moments are all calculated in the SIPP except for the external moments and contact gaps.

Table A.10 shows the decomposition of the steady-state difference in employment gaps

across the alternative calibrations without wage dispersion. Allowing firms to pay lower wages to Black workers is important for the large negative policy effects at large firms in the calibrations with discrimination and incumbency bias. When all firms are required to pay equal wages, the employment gap improves at large firms while worsening more severely at small firms. The second column shows an intermediate case, where large firms must pay equal wages but small firms are allowed to pay competitive wages. This tilts the responsiveness of the employment gap away from small firms and towards large firms, though small firms still contribute more overall. Finally, in the simplified calibration that does not explicitly model the channels of discrimination, large firms still dominate the negative change in the employment gap, though the magnitude is smaller than the results with wage dispersion, shown in Table 8. Overall, this suggests that wage dispersion amplifies the employment response at large firms.

Table A.10: Steady state comparison without wage dispersion

	Baseline	No wage dispersion	Only wage disp. at small firms	No wage disp, exog. bias
Δ White employment rate	-1.49	-1.49	-1.49	-1.49
Large	-1.04	-1.40	-1.21	-1.07
Small	-0.45	-0.08	-0.28	-0.42
Δ Employment gap	-1.21	-0.52	-0.81	-0.14
Large	-1.21	0.45	-0.34	-0.23
Small	0.01	-0.98	-0.46	0.09

This table shows the change in the white employment rate and racial employment gap between steady states. The first column uses the baseline calibration and the subsequent columns report results when I remove firm's ability to pay different wages for workers with the same productivity. The first scenario removes all wage discrimination. The second removes wage discrimination just at large firms. The last removes wage discrimination in the model with exogenously lower matching rates for Black workers. The units are percentage points. The low productivity value is chosen such that the change in white employment in the first row is the same across calibrations.